Identification and Optimal Reserve Prices in English Auctions, with an Application to Art

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Abstract

We develop a partial-identification approach for English auctions with private values, auction-level heterogeneity unobserved by the analyst, and a potentially unknown number of bidders. Our framework permits restrictions on dependence among bidders' valuations away from the extreme cases of independent private values and perfect correlation, yielding sharp and informative bounds on the distribution of the highest valuation and the seller's expected profit. We formulate reserve-price choice as a minimax-regret problem and derive tractable solutions to two relaxations of this problem. We apply our method to a new dataset covering more than 3,500 lots auctioned by Christie's and Sotheby's. For modern art sold in New York City with presale estimates between $1 million and $10 million, we find that setting the reserve price 37 percent above the sellers projected value from retaining an unsold lot increases expected profit by approximately $150,000 to $489,000 per lot, equivalent to 27 to 87 percent of average realized profit.

Keywords: nonparametric identification, ascending auctions, partial identification, minimax regret, unknown number of bidders, correlated values, art auctions.

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1 Introduction

English auctions, or open ascending-bid auctions, are widely used to sell goods ranging from used cars, timber, and agricultural products to real estate, rare wine, and fine art. Their empirical analysis, however, presents two important challenges. First, bidders may possess information about the object that is unavailable to the analyst, which induces an unknown dependence structure among their valuations. Second, the number of bidders may itself be uncertain, because attendance, active participation, and the opportunity to submit a bid need not coincide in an ascending auction. Consequently, the distribution of the highest valuation, the sellers expected profit, and consumer surplus generally cannot be point identified from observed bids alone. Existing approaches address these challenges either by imposing strong restrictions on the joint distribution of valuations, which may be difficult to justify in many applications, or by relying on additional information, such as reserve prices, realized seller profits, or multiple losing bids, that is often unavailable to the analyst.

In this paper, we develop a partial identification approach that addresses these challenges and apply it to the seller's problem of choosing a reserve price. We study English auctions with symmetric private values and assume that, conditional on a sale, the transaction price equals the maximum of the reserve price and the second-highest valuation. Under this assumption, the only unidentified input into the seller's expected profit is the marginal distribution of the highest valuation. Observed bids alone do not identify this distribution because the auction ends before the highest bidder reveals their valuation. To make progress, we assume that bidders' valuations are independent and identically distributed conditional on the information commonly observed by the bidders but not by the analyst. The limiting cases of this framework, corresponding to independent private values perfectly correlated values, produce non-trivial but wide identified sets. By restricting dependence among valuations away from these limits, we obtain sharp, informative, and computationally simple bounds on the marginal distribution of the highest valuation and, consequently, on the seller's expected profit.

Because the distribution of valuations is only partially identified, the seller faces ambiguity about the expected profit function. We address this ambiguity using a minimax-regret criterion. Since the sharp identified set for expected profit is analytically intractable, we formulate two tractable relaxations of the resulting decision

problem and characterize their solutions. We evaluate the resulting reserve prices in empirically calibrated simulations, comparing them with max-min and zero-reserve benchmarks. We show that the minimax-regret reserves can generate substantial gains in expected profit relative to both alternatives. We also characterize conditions under which the minimax-regret and max-min reserve prices coincide.

We apply our method to art auctions conducted by Christies and Sothebys. Although auction houses publish final sale prices and detailed information about individual lots, they disclose relatively little about the bidding process: historical bid sequences, bidder identities, and the number of active bidders are generally unavailable. To overcome this limitation, we construct a new dataset from publicly available livestream recordings of auctions held between 2020 and 2024, covering more than 3,500 lots. For each lot, we recover the bid sequence and bounds on the number of bidders and collect detailed characteristics, including presale estimates, auction location, art category, artist, period, provenance, and condition. For modern art sold in New York City with presale estimates between $1 million and $10 million, our procedure recommends a reserve price 37 percent above the sellers projected value from retaining an unsold lot. The resulting increase in expected profit ranges from approximately $150,000 to $489,000 per lot, corresponding to 27 to 87 percent of average realized profit.

Our identification analysis contributes to the literature on nonparametric identification in ascending auctions.[1] Assuming independent private values, a known number of bidders, and equilibrium play, Athey and Haile (2002) established point identification of the marginal distribution of bidders' values. In this setting, the distribution of any order statistic from an i.i.d. sample identifies the parent distribution. Haile and Tamer (2003) relaxed the equilibrium play assumption, allowing for minimal bid increments and jump bidding, and obtained bounds for the marginal distribution of values, expected profit, and optimal reserve price. Recently, Chesher and Rosen (2017) and Molinari (2020) characterized the corresponding sharp identified sets using tools from random set theory.

Our results are most closely related to Aradillas-López, Gandhi, and Quint (2013), who studied ascending auctions with auction-level unobserved heterogeneity. The authors noted that, when the transaction price is the maximum of the reserve price and

[1]We focus on ascending auctions with private values and refer the reader to Athey and Haile (2007) and Hortaçsu and Perrigne (2021) for a discussion of the broader auction literature.

the second-highest valuation, the seller's expected profit and bidder surplus depend only on the marginal distributions of the two highest valuations. Assuming that valuations are symmetric and positively dependent, they derived analytical bounds corresponding to the limiting cases of independent private values (IPV) and perfectly correlated values (PCV) and proposed tightening those bounds by using exogenous variation in the number of bidders. We instead impose a stronger but transparent restriction on the dependence among valuations that bounds it away from both IPV and PCV, leading to tighter bounds (Theorems 1–2). Our approach is particularly useful when neither limiting case provides a credible approximation or when exogenous variation in the number of bidders is unavailable. Computing our bounds amounts to solving two generalized moment problems: maximizing and minimizing the expectation of a convex function of a random variable subject to moment constraints. By exploiting the analytical properties of this function, we reduce both problems to univariate convex optimization, making the bounds easy to compute in practice.

A separate literature studies uncertainty about the number of bidders. Under independent private values, Song (2004) identifies the distribution of valuations from the conditional distribution of one order statistic given another, without observing the number of bidders. Marra (2020) instead exploits stochastic orderings between adjacent order statistics. With auction-level unobserved heterogeneity, existing approaches generally require additional information. Freyberger and Larsen (2022) use observations of the sellers profit; Mbakop (2017) and Luo and Xiao (2023) use multiple bids and impose finite support on the unobserved heterogeneity; and Hernández, Quint, and Turansick (2020) use information about the distribution of the number of bidders. We instead develop a partial-identification approach that requires only observed or conservatively specified lower and upper bounds on the number of bidders. The approach applies both with and without unobserved heterogeneity and exploits the monotonicity of the bounds on the distribution of the highest valuation with respect to the number of bidders (Theorems 3–4).

The reserve price analysis is related to work on decision-making under ambiguity. In the IPV setting, Aryal and Kim (2013) proposed the max-min reserve, obtained by maximizing the lower bound on expected profit, motivating it from a robust Bayesian perspective. Jun and Pinkse (2024) proposed choosing the reserve price that maximizes the profit function under the maximum entropy distribution within the identified set, arguing that this approach utilizes more of the available identifying

information and typically leads to higher reserves than max-min. The minimax-regret criterion originated in Savage (1951) and has been influential in econometrics following Manski (2004). It has also been used to study monopoly pricing under ambiguity by Bergemann and Schlag (2008) and Handel, Misra, and Roberts (2013). To the best of our knowledge, it has not been considered in the auction literature.

In our setting, ambiguity concerns the distribution of the highest valuation. Because its sharp identified set is analytically intractable, we formulate two relaxations of the minimax-regret problem based on outer identified sets. The first allows any distribution function lying between the available lower and upper bounds. The second additionally requires the density ratio of the highest to the second-highest valuation to be nondecreasing, thus imposing a strong form of stochastic dominance. We solve both relaxations analytically and obtain reserve-price recommendations that are straightforward to compute (Theorems 5–6). We further show that, when the admissible range of dependence among valuations is sufficiently wide, the minimax-regret and max-min reserve prices coincide. Using our tighter bounds on expected profit, we find that both criteria can perform well in practice.

Finally, our empirical application contributes to the literature on art and wine auctions, including Ashenfelter (1989); Ashenfelter and Graddy (2003); Beggs and Graddy (2009); Ashenfelter and Graddy (2011); McAndrew, Smith, and Thompson (2012), and Marra (2020). A common finding in the literature is that the reserve prices are suboptimally low. We confirm this finding empirically using a large new dataset of art auctions and propose a theoretically justified way to choose a higher reserve price in practice.

The rest of the paper is organized as follows. Section 2 gives a formal setup and presents the main identification results; Section 3 discusses the seller's problem under ambiguity; Section 4 briefly discusses estimation and inference; Section 5 illustrates the proposed approach with simulated data; Section 6 discusses data collection and applies the proposed methodology to art auctions; and Section 7 concludes.

2 Identification in English Auctions

2.1 Auction Environment and Bidding Behavior

Consider an open ascending (i.e., English) auction with N bidders. Let $V_1, \ldots, V_N \in \mathcal{V}$ denote the bidders' values, $V_{1:N}, \ldots, V_{N:N}$ denote the corresponding order statistics, from smallest to largest, and $F_{j:N}$ denote the distribution of $V_{j:N}$, conditional on N. We model the bidders' values as continuous random variables. Throughout, we implicitly condition on all auction-level covariates that are observed by the analyst. In this section, we assume that N is observed, so the main object of interest is the seller's expected profit, conditional on N. Following Aradillas-López, Gandhi, and Quint (2013), we impose two standard assumptions.

Assumption 2.1. *Bidders have symmetric private values.*

That is, the bidders know their values and are observationally indistinguishable to the analyst. Formally, the assumption requires that the joint distribution of $(V_1, \ldots, V_N)$ be exchangeable.

Assumption 2.2. *In each auction, the transaction price is the greater of the reserve price and the second-highest valuation.*

With private values, bidding one's valuation is a weakly dominant strategy in an ascending auction without minimal bid increments, so Assumption 2.2 holds exactly under equilibrium play. In many settings, it holds approximately if bidders do not "jump bid" at the end of the auction, which is easy to verify empirically. This assumption can be relaxed as in Haile and Tamer (2003), as discussed in Section 7.

Let $v_0 \in \mathcal{V}$ be the value of the unsold good to the seller, and $r \in \mathcal{V}$ the reserve price. Under Assumption 2.2, the seller's realized profit is

$$\begin{aligned} \mathcal{P}(r) &= (r - v_0) \cdot \mathbf{1}(V_{N:N} \geq r, V_{N-1:N} < r) + (V_{N-1:N} - v_0) \cdot \mathbf{1}(V_{N-1:N} \geq r) \\ &= \max(r, V_{N-1:N}) - (r - v_0)\mathbf{1}(V_{N:N} < r) - v_0. \end{aligned} \tag{1}$$

Thus, the expected profit, conditional on N, is

$$\pi_N(r) \equiv \mathbb{E}[\mathcal{P}(r) \mid N] = \int_0^{+\infty} \max(r, v) dF_{N-1:N}(v) - v_0 - (r - v_0)F_{N:N}(r). \tag{2}$$

Notably, the expected profit only depends on the marginal distributions of the two highest values.[2] If the reserve prices are not binding, the distribution $F_{N-1:N}$ is nonparametrically identified, so the only unknown component is $F_{N:N}$. Since the auction ends before the highest bidder reveals their value, the data do not provide any information about $F_{N:N}$ beyond $F_{N:N}(v) \leq F_{N-1:N}(v)$, for all $v \in \mathcal{V}$. Thus, further assumptions are required to make progress. We consider the following setting.

Assumption 2.3. *Conditional on N, bidders' values satisfy $V_j = g(U, \varepsilon_j)$, for $j \in \{1, \ldots, N\}$, where $\varepsilon_j \in \mathbf{R}$ are i.i.d. random variables, $U \in \mathbf{R}^{d_U}$ is a random vector, independent of the ε_j-s, and $g : \mathbf{R}^{d_U} \times \mathbf{R} \to \mathcal{V}$ is measurable.*

That is, conditional on the number of bidders N and auction-level information U, the values $V_1, \ldots, V_N$ are i.i.d. The distributions of U and ε_j, as well as the function $g(\cdot)$, may depend on N. In art auctions, for example, U may represent a projected future resale value of the lot, and ε_j individual consumption value. In addition to the values being symmetric, Assumption 2.3 implies a form of positive dependence: for any measurable function $h : \mathcal{V} \to \mathbf{R}$ such that $\mathbb{E}[h(V_j)^2 \,|\, N] < \infty$ (almost surely) and any pair $i \neq j$, it follows that $\mathbb{C}ov(h(V_i), h(V_j) \,|\, N) \geq 0$.[3] At the same time, the assumption allows the strength of the dependence between the values to range from independent private values (IPV) to perfectly correlated values (PCV).

Assumption 2.3 implies simple bounds on $F_{N:N}$ in terms of the observed $F_{N-1:N}$. Consider a function $\phi_N : [0,1] \to [0,1]$ defined implicitly as

$$u = N\phi_N(u)^{N-1} - (N-1)\phi_N(u)^N. \tag{3}$$

Given an i.i.d. sample $Z_1, \ldots, Z_N$, this function maps the CDF of $Z_{N-1:N}$ to that of Z_i, that is, $P(Z_i \leq z) = \phi_N(P(Z_{N-1:N} \leq z))$.

Proposition 1. *Let Assumption 2.3 hold and $\phi_N : [0,1] \to [0,1]$ be defined in (3). Then, for all $N \geq 2$, the function $u \mapsto \phi_N(u)^N$ is convex and*

$$\phi_N(F_{N-1:N}(v))^N \leq F_{N:N}(v) \leq F_{N-1:N}(v), \tag{4}$$

for all $v \in \mathcal{V}$. The bounds are sharp uniformly over $v \in \mathcal{V}$. The lower bound corresponds to the IPV and the upper bound to PCV.

[2] The same applies to expected surplus, see Aradillas-López, Gandhi, and Quint (2013).

[3] The stated inequality follows from the law of total covariance, $\mathbb{C}ov(h(V_i), h(V_j) \,|\, N) = \mathbb{V}ar(\mathbb{E}[h(V_i) \,|\, U, N] \,|\, N) \geq 0$.

Aradillas-López, Gandhi, and Quint (2013) obtained the same bounds under a weaker notion of positive dependence.[4] They noted that, since Assumption 2.3 does not restrict the strength of dependence between the values, the bounds in (4) are typically wide. In particular, the implied lower bound on the expected profit, obtained by plugging in $F_{N-1:N}$ in place of $F_{N:N}$ in Equation (2), is monotonically decreasing in r, for $r \geq v_0$, and thus is not informative about the optimal reserve price. To make progress, the authors tightened the bounds by leveraging exogenous variation over N.

We take a different approach. Since the extreme cases of IPV and PCV may be implausible, it is natural to further restrict the dependence between the values. To this end, note that the bounds in (4) solve the following generalized moment problems:

$$\begin{aligned} &\min_P\{\mathbb{E}_P[\phi_N(X)^N] : \mathbb{E}_P[X] = \mu, P(X \in [0,1]) = 1\} = \phi_N(\mu)^N, \\ &\max_P\{\mathbb{E}_P[\phi_N(X)^N] : \mathbb{E}_P[X] = \mu, P(X \in [0,1]) = 1\} = \mu, \end{aligned} \tag{5}$$

where X represents the random variable $P(V_{N-1:N} \leq v \,|\, U, N)$, conditional on N, $\mu = F_{N-1:N}(v)$, and the minimum and maximum are taken over all Borel probability measures P. The lower bound corresponds to Jensen's inequality and the upper bound to the so-called Edmundson-Madansky inequality for the convex function $u \mapsto \phi_N(u)^N$ on $u \in [0,1]$ (Edmundson, 1957; Madansky, 1959). Without further restrictions on the distribution of $P(V_{N-1:N} \leq v \,|\, U, N)$, conditional on N, these bounds are sharp.

To tighten the bounds, we further restrict the variance of $P(V_{N-1:N} \leq v \,|\, U, N)$, conditional on N, denoted

$$D_N(v) = \mathbb{E}[(P(V_{N-1:N} \leq v \,|\, U, N) - F_{N-1:N}(v))^2 \,|\, N]. \tag{6}$$

Since $P(V_{N-1:N} \leq v \,|\, U, N)$ is a random variable supported within $[0,1]$, $D_N(v)$ takes values in $[0, F_{N-1:N}(v)(1 - F_{N-1:N}(v))]$, for each $v \in \mathcal{V}$. The lower bound corresponds to IPV and the upper bound to PCV.[5] That is, $D_N(v)$ quantifies the contribution of U to $V_{N-1:N}$, and bounding it away from the extremes allows us to control the

[4] The authors assume that the values are exchangeable and the function $f(k) = P(V_i \leq v \,|\, \#\{j \neq i : V_j \leq v\} = k)$ is nondecreasing in k. Lemma 1 in the aforementioned paper shows that conditional independence implies their notion of positive dependence.

[5] If $D_N(v) = 0$, for all v, then $P(V_{N-1:N} \leq v \,|\, U, N) = F_{N-1:N}(v)$, almost surely, for all v. Under Assumption 2.3, this implies that the conditional distribution of each bidder's value does not depend on U, corresponding to IPV conditional on N. If $D_N(v) = F_{N-1:N}(v)(1 - F_{N-1:N}(v))$, for all v, then it must be that $P(V_{N-1:N} \leq v \,|\, U, N) \in \{0, 1\}$, for all v, which, under Assumption 2.3 implies that each V_j is a deterministic function of U, corresponding to PCV conditional on N.

dependence between the values.

To further interpret $D_N(v)$, suppose that $V_{N-1:N} = g(U, \varepsilon_{N-1:N})$. Let $\tilde{\varepsilon}_{N-1:N}$ be an independent copy of $\varepsilon_{N-1:N}$, also independent of U, conditional on N, and $\tilde{V}_{N-1:N} = g(U, \tilde{\varepsilon}_{N-1:N})$. By the law of iterated expectations,

$$\begin{aligned}
\mathbb{E}[P(V_{N-1:N} \leq v \,|\, U, N)^2 \,|\, N] &= \mathbb{E}[P(V_{N-1:N} \leq v \,|\, U, N) \cdot P(\tilde{V}_{N-1:N} \leq v \,|\, U, N) \,|\, N] \\
&= \mathbb{E}[P(V_{N-1:N} \leq v, \tilde{V}_{N-1:N} \leq v \,|\, U, N) \,|\, N] \\
&= P(V_{N-1:N} \leq v, \tilde{V}_{N-1:N} \leq v \,|\, N) \\
&= C_N(F_{N-1:N}(v), F_{N-1:N}(v)),
\end{aligned} \tag{7}$$

where $C_N : [0,1]^2 \to [0,1]$ is the copula of $(V_{N-1:N}, \tilde{V}_{N-1:N})$, conditional on N.[6] Thus, we can generally express

$$D_N(v) = C_N(F_{N-1:N}(v), F_{N-1:N}(v)) - F_{N-1:N}(v)^2. \tag{8}$$

The copula function C_N captures the dependence between the second-highest values in two auctions with N bidders, with the same auction-level information and independent draws of the bidders. If the data contains pairs of auctions of identical goods (e.g., iPads or used cars) with independent draws of the bidders, the copula C_N, and therefore the variance D_N can be nonparametrically identified. Otherwise, the analyst may impose bounds on D_N. To accommodate both scenarios, we impose the following assumption.

Assumption 2.4. *(i) The map $e \mapsto g(U, e)$ in Assumption 2.3 is weakly increasing, U-a.s.;*

(ii) The function $D_N(v)$ in (6), satisfies $\underline{D}_N(v) \leq D_N(v) \leq \overline{D}_N(v)$, for all $v \in \mathcal{V}$, where

$$\begin{aligned}
\underline{D}_N(v) &= \underline{C}_N(F_{N-1:N}(v), F_{N-1:N}(v)) - F_{N-1:N}(v)^2 \\
\overline{D}_N(v) &= \overline{C}_N(F_{N-1:N}(v), F_{N-1:N}(v)) - F_{N-1:N}(v)^2,
\end{aligned}$$

for symmetric copula functions $\underline{C}_N, \overline{C}_N : [0,1]^2 \to [0,1]$, known or identified, satisfying: $u_1 u_2 \leq \underline{C}_N(u_1, u_2) \leq \overline{C}_N(u_1, u_2) \leq \min(u_1, u_2)$, for all $u_1, u_2 \in [0,1]$.

[6] The copula function of random variables X and Y with a joint CDF F_{XY} is defined as a function $C : [0,1]^2 \to [0,1]$ satisfying $F_{XY}(x,y) = C(F_X(x), F_Y(y))$. If X and Y are continuous, $C(u_1, u_2) = P(F_X(X) \leq u_1, F_Y(Y) \leq u_2)$, where $F_X(X), F_Y(Y)$ are marginally Uniform $[0,1]$.

(iii) *Let* $(U_1, U_2) \in \mathbf{R}^2$ *be random variables with a joint distribution* $C \in \{\underline{C}_N, \overline{C}_N\}$. *Then,* (U_1, U_2) *satisfy* $P(U_1 \leq u \,|\, U_2 = u) \geq P(U_1 \leq u \,|\, U_2 > u)$, *for all* $u \in (0,1)$, *and* $u \mapsto P(U_1 \leq u \,|\, U_2 \leq u)$ *is nondecreasing.*

Condition (i) is natural and ensures validity of the copula representation in (8). Condition (ii) requires that the copula functions $\underline{C}_N, \overline{C}_N$ agree with Assumption 2.3: they must be symmetric and lie between the independence copula, $C_0(u_1, u_2) = u_1 u_2$, and the comonotone copula, $C_1(u_1, u_2) = \min(u_1, u_2)$, which maintains IPV and PCV as the marginal cases and rules out negative dependence between $V_{N-1:N}$ and $\tilde{V}_{N-1:N}$. Condition (iii) is an additional shape restriction which will ensure that the bounds on $F_{N:N}$ derived in Theorem 1 below are valid CDFs. It is satisfied by a variety of copulas capturing non-negative dependence, including the independence and comonotone copulas. A sufficient condition for (iii) is $\mathbb{C}ov(h_1(V_{N-1:N}), h_2(\tilde{V}_{N-1:N})) \geq 0$, for any nondecreasing functions $h_1, h_2 : \mathcal{V} \to \mathbf{R}$.

When $D_N(v)$ is not identified, suitable bounds can be specified as follows. Let $C_\rho : [0,1]^2 \to [0,1]$ be a symmetric copula function, parametrized by $\rho \in [0,1]$ such that: (i) For each $u \in (0,1)$, $\rho \mapsto C_\rho(u,u)$ is nondecreasing; $C_0(u_1, u_2) = u_1 u_2$ and $C_1(u_1, u_2) = \min(u_1, u_2)$; (ii) $(C_\rho(u,u))' \geq \max(2\frac{u - C_\rho(u,u)}{1-u}, \frac{C_\rho(u,u)}{u})$. These conditions exactly match conditions (ii) and (iii) in Assumption 2.4.[7] With this parametrization, $\rho \in [0,1]$ measures the dependence between $V_{N-1:N}$ and $\tilde{V}_{N-1:N}$ coming only through U. Under additive model, $V_j = U + \varepsilon_j$, ρ measures the share of the variation in the price attributable to U. For example, $\rho = 0.8$ means that 80% of the variation in transaction prices is attributable to common auction-level factors, such as future resale prospects, and the remaining 20% to idiosyncratic factors, such as bidders' personal consumption values.

The analyst may specify $\underline{C}_N = C_{\underline{\rho}}$ and $\overline{C}_N = C_{\overline{\rho}}$ for some $0 \leq \underline{\rho} \leq \overline{\rho} \leq 1$. Setting $\underline{\rho} > 0$ ensures a non-trivial common component, and $\overline{\rho} < 1$ ensures a non-trivial idiosyncratic component.[8] One may fix suitable values of $\underline{\rho}, \overline{\rho}$ based on their prior knowledge or calibration, or vary these values starting from a benchmark specification (e.g., $\underline{\rho} = \overline{\rho} = 0$) and perform sensitivity analysis (Masten and Poirier, 2020). Since

[7] In practice, it suffices to choose a function $C_\rho(u,u)$ that interpolates between u and u^2. Lemma C.4 in Appendix verifies that natural choices, such as $C_\rho(u,u) = \rho u + (1-\rho)u^2$, $C_\rho(u,u) = u^{2-\rho}$, or $C_\rho(u,u)$ the bivariate Gaussian copula with correlation $\rho \in [0,1]$, satisfy the stated assumptions. Other copulas can be used when suitably reparametrized (see Table 4.1. in Nelsen, 2006).

[8] Setting $\underline{\rho} = \overline{\rho}$ pins down the CDF of $(V_{N-1:N}, \tilde{V}_{N-1:N})$ along the diagonal (v,v), leaving it completely unspecified otherwise. In such case, without further assumptions, the CDF of the highest value $F_{N:N}(v)$ is still only partially identified.

U represents auction-level information that remains after conditioning on covariates, lower values of $\overline{\rho}$ may be more appropriate with richer covariate data. We discuss the calibration of $\underline{\rho}, \overline{\rho}$ in detail in Section 6.

2.2 CDF of the Highest Value and Expected Profit

Under Assumptions 2.3 and 2.4, pointwise sharp bounds on $F_{N:N}$ can be obtained by solving generalized moment problems similar to the ones in (5) with an extra constraint of the form $c_1 \leq \mathbb{V}ar(X) \leq c_2$. Using a geometric argument from Kemperman (1968), we show that both bounds can be computed by solving univariate convex optimization problems.

Theorem 1. *Let Assumptions 2.3 and 2.4(i)(ii) hold. Then, for each $v \in \mathcal{V}$,*

$$\underline{\psi}_N(F_{N-1:N}(v)) \leq F_{N:N}(v) \leq \overline{\psi}_N(F_{N-1:N}(v)), \tag{9}$$

where

$$\begin{aligned}
\underline{\psi}_N(\mu) &= \min_{s \in [\frac{c_1}{1-\mu},\, \mu]} \left\{ \phi_N(\mu - s)^N \tfrac{c_1}{c_1 + s^2} + \phi_N\left(\mu + \tfrac{c_1}{s}\right)^N \tfrac{s^2}{c_1 + s^2} \right\}; \\
\overline{\psi}_N(\mu) &= \max_{s \in \left[\mu - \frac{c_2}{1-\mu},\, \mu + \frac{c_2}{\mu}\right]} \left\{ \mu - \tfrac{\mu(1-\mu) - c_2}{s(1-s)} (s - \phi_N(s)^N) \right\},
\end{aligned}$$

with $\mu = F_{N-1:N}(v)$, $c_1 = \underline{C}_N(\mu, \mu) - \mu^2$, and $c_2 = \overline{C}_N(\mu, \mu) - \mu^2$. The minimization problem is strictly convex, and the maximization problem is strictly concave. Moreover, if Assumption 2.4(iii) holds, the bounds are valid CDFs.

With $\underline{D}_N(v) = 0$ and $\overline{D}_N(v) = F_{N-1:N}(v)(1 - F_{N-1:N}(v))$ (or $\underline{\rho} = 0, \overline{\rho} = 1$), the bounds in Theorem 1 coincide with those in Proposition 1, because they solve the moment problems in (5). Otherwise, the bounds in Theorem 1 are strictly tighter. If only the transaction price is observed in each auction, the bounds are sharp pointwise in $v \in \mathcal{V}$. Under Assumption 2.4(iii), the bounds are monotonically increasing, so they are plausibly sharp in the first-order stochastic dominance sense, although this property is hard to establish formally. Notably, certain CDFs within the bounds may not belong to the sharp identified set for $F_{N:N}$: for example, they may be inconsistent with the conditional i.i.d. structure in Assumption 2.3, or contain atoms. Although the sharp identified set for $F_{N:N}$ is hard to describe tractably, the proposed bounds are typically informative, as we demonstrate below.

Theorem 1 implies the following bounds on the expected profit.

Theorem 2. *Let Assumptions 2.1–2.4 hold. Let* $\underline{\psi}_N(F_{N-1:N}(v))$ *and* $\overline{\psi}_N(F_{N-1:N}(v))$ *denote the lower and upper bounds on* $F_{N:N}(v)$ *obtained in Theorem 1. Then,*

$$\begin{aligned}\pi_N(r) &\geq \int_0^\infty \max\{r,v\}dF_{N-1:N}(v) - v_0 - \overline{\psi}_N(F_{N-1:N}(r))(r-v_0),\\ \pi_N(r) &\leq \int_0^\infty \max\{r,v\}dF_{N-1:N}(v) - v_0 - \underline{\psi}_N(F_{N-1:N}(r))(r-v_0),\end{aligned}$$

for all $r \geq v_0$.

As in Theorem 1, the bounds are sharp pointwise and plausibly sharp uniformly.[9] However, many profit functions within the bounds cannot be attained under the stated assumptions, which will be important for studying profit-maximizing reserve prices.

2.3 Unknown Number of Bidders

In English auctions, the number of active bidders is often unknown to the analyst, or even to the seller. For example, it may happen that an early "jump bid" from one of the bidders prevented the other potential bidders from making a bid, thus effectively excluding them from participation. Below, we extend the analysis assuming that bounds on the number of bidders are available.

Assumption 2.5. *The number of bidders* N *is not observed but it satisfies* $\underline{N} \leq N \leq \overline{N}$*, almost surely, for some observed* $2 \leq \underline{N} \leq \overline{N}$.

The natural objective in this context is the expected profit, conditional on $(\underline{N}, \overline{N})$,

$$\pi(r) \equiv \mathbb{E}[\pi_N(r) \,|\, \underline{N}, \overline{N}] = \int_0^\infty \max(r,v)dF_2(v) - v_0 - F_1(r)(r-v_0), \tag{10}$$

where $\pi_N(r)$ is as in (2), $F_2(v) \equiv \mathbb{E}[F_{N-1:N}(v) \,|\, \underline{N}, \overline{N}]$ and $F_1(v) \equiv \mathbb{E}[F_{N:N}(v) \,|\, \underline{N}, \overline{N}]$. If the reserve prices are not binding and Assumption 2.2 holds, the distribution F_2 is identified from the data, by the law of iterated expectations. As before, the main challenge is in obtaining informative bounds on F_1. To this end, we impose the following assumption.

[9] The bounds on expected profit for r below v_0 can be obtained by switching the roles of $\underline{\psi}_N(\cdot)$ and $\overline{\psi}_N(\cdot)$, but are irrelevant because it is never optimal to set the reserve price below v_0 (which follows from the FOCs for maximizing $\pi_N(r)$).

Assumption 2.6. *Let $(U, \varepsilon_1, \ldots, \varepsilon_N)$ be as defined in Assumption 2.3. $(U, \varepsilon_1, \ldots, \varepsilon_N)$ is statistically independent of $(\underline{N}, \overline{N})$, conditional on N.*

That is, the observed bounds on the number of bidders are not informative about their values, conditional on N. This assumption concerns solely construction of the bounds and does not restrict the dependence between $(U, \varepsilon_1, \ldots, \varepsilon_N)$ and N.

To simplify notation, below we keep conditioning on $(\underline{N}, \overline{N})$ in *all* probabilities and expectations implicit. Under Assumption 2.6, we can express

$$\begin{aligned} F_1(v) &= \textstyle\sum_{j=\underline{N}}^{\overline{N}} P(N=j)\mathbb{E}[\phi_j(P(V_{N-1:N} \le v \,|\, U, N=j))^j \,|\, N=j] \\ F_2(v) &= \textstyle\sum_{j=\underline{N}}^{\overline{N}} P(N=j)\mathbb{E}[P(V_{N-1:N} \le v \,|\, U, N=j) \,|\, N=j]. \end{aligned}$$

Here, obtaining sharp bounds on $F_1(v)$ amounts to solving more complex optimization problems than the ones in (5): Instead of choosing a single best- or worst-case distribution of $P(V_{N-1:N} \le v \,|\, U, N=j)$, we must choose several such distributions simultaneously, and the corresponding best- or worst-case distribution of N. However, we show that, for each $t \in [0,1]$, the function $j \mapsto \phi_j(t)^j$ is weakly increasing, so the sharp bounds take a simple form.

Proposition 2. *Let Assumptions 2.5 and 2.6 hold. Then, for all $v \in \mathcal{V}$,*

$$\phi_{\underline{N}}(F_2(v))^{\underline{N}} \le F_1(v) \le F_2(v).$$

Moreover, the bounds are sharp uniformly over $v \in \mathcal{V}$.

The lower bound corresponds to a DGP for which $P(N = \underline{N}) = 1$, and the values are i.i.d., and the upper bound can be attained by a DGP for which $P(N = n) = 1$, for some fixed $n \in \{\underline{N}, \ldots, \overline{N}\}$, and the values are perfectly correlated. For the same reasons as in Proposition 1, such bounds are generally very wide, which motivates imposing further restrictions. To obtain tighter bounds while maintaining tractability, we leave the distribution of N completely unspecified and restrict the unconditional variance

$$D(v) = \mathbb{E}[(P(V_{N-1:N} \le v \,|\, U, N) - F_2(v))^2].$$

As we show in the proof of Theorem 3 below, sharp bounds on F_1 under additional restrictions on $D(v)$ correspond to DGPs for which either $P(N = \underline{N}) = 1$ or $P(N =$

$\overline{N}) = 1$. Thus, restricting $D(v)$ amounts to restricting $D_{\underline{N}}(v)$ or $D_{\overline{N}}(v)$ separately. As a result, we obtain direct analogs of Theorems 1 and 2 with unknown N as follows.

Theorem 3. *Let Assumptions 2.5 and 2.6 hold, and Assumptions 2.4(i)(ii) hold for $N \in \{\underline{N}, \overline{N}\}$. Then, for each $v \in \mathcal{V}$,*

$$\underline{\psi}_{\underline{N}}(F_2(v)) \leq F_1(v) \leq \overline{\psi}_{\overline{N}}(F_2(v)),$$

where $\underline{\psi}_N(\cdot)$ and $\overline{\psi}_N(\cdot)$ are as defined in Theorem 1. If, additionally, Assumption 2.4(iii) holds, for $N \in \{\underline{N}, \overline{N}\}$, the stated bounds on $F_1(\cdot)$ are valid CDFs.

Theorem 4. *Let Assumptions 2.1, 2.2, 2.5, 2.6 hold, and Assumption 2.4 hold for $N \in \{\underline{N}, \overline{N}\}$. Then, for all $r \geq v_0$,*

$$\begin{aligned}
\pi(r) &\geq \int_0^\infty \max\{r,v\} dF_2(v) - v_0 - \overline{\psi}_{\overline{N}}(F_2(r))(r - v_0); \\
\pi(r) &\leq \int_0^\infty \max\{r,v\} dF_2(v) - v_0 - \underline{\psi}_{\underline{N}}(F_2(r))(r - v_0),
\end{aligned}$$

where $\underline{\psi}_{\underline{N}}(\cdot)$ and $\overline{\psi}_{\overline{N}}(\cdot)$ are the bounding functions defined in Theorem 1.

As in Theorems 1 and 2, the bounds are pointwise sharp. One may wish to impose additional restrictions on the distributions of $V_{N-1:N}$, conditional on N, such as $F_{N-1:N} \,|\, N = n \preceq_{FOSD} F_{N-1:N} \,|\, N = n+1$, i.e., that attracting more bidders yields a higher expected profit for the seller (Bulow and Klemperer, 1996). However, the proof of Theorem 3 shows that if the distribution of N is left unconstrained, adding such stochastic dominance restrictions does not change the resulting bounds on F_1. Restricting the distribution of N is possible in our framework, but solving the resulting optimization problems requires substantially different techniques. Thus, we leave it for further research.

3 Choosing Reserve Prices Under Ambiguity

Consider the decision problem of a seller who sets a reserve price $r \in \mathcal{R}_0$ prior to an auction. For the sake of discussion in this section, suppose that the number of prospective bidders N is known. Under Assumption 2.2, if the reserve price is set at r, the payoff of the seller is $\mathcal{P}(r) = \max(r, V_{N-1:N}) - (r - v_0)\mathbf{1}(V_{N:N} < r) - v_0$.

The distribution of $\mathcal{P}(r)$ depends on the joint distribution of the two largest values, $(V_{N-1:N}, V_{N:N})$. Let $F \in \mathcal{F}_0$ denote the joint distribution of $(V_{N-1:N}, V_{N:N})$, and $J(\cdot\,; r, F)$ denote the corresponding distribution of $\mathcal{P}(r)$. Following the standard terminology, we refer to F as a *state*, and $J(\cdot\,; r, F)$ as a *lottery*.

The seller has a preference ordering over lotteries and chooses between them. If the state F was known, under standard assumptions on the seller's preference ordering, it would be consistent with maximizing expected profit, $\pi_N(r) = \mathbb{E}[\mathcal{P}(r) \,|\, N]$. In our setting, F is only partially known: the data from past auctions allow us to identify the marginal distribution of $V_{N-1:N}$ and bound the marginal distribution of $V_{N:N}$, as we discussed in Section 2, but do not reveal the joint distribution of these random variables. Thus, the seller faces ambiguity.

Each reserve price $r \in \mathcal{R}_0$ induces a set of lotteries $\{J(\cdot\,; r, F) : F \in \mathcal{F}_0\}$. In the standard terminology, $\mathcal{R}_0$ defines the *menu of acts* available to the seller. Existing approaches to modeling decisions under ambiguity assume that the decision-maker has a preference ordering over acts. If such preference imposes prior beliefs by implicitly assigning different likelihoods to different states, it may be consistent with maximizing a subjective expected utility (Anscombe and Aumann, 1963) or, under ambiguity aversion, worst-case subjective expected utility (Gilboa and Schmeidler, 1989). If the preference does not impose any prior beliefs and exhibits ambiguity aversion, it may be consistent with maximizing the worst-case expected utility (Wald, 1950) or minimizing the worst-case regret (Savage, 1951). See Stoye (2011) for a detailed discussion. In what follows, we discuss some of the existing approaches in more detail and make a case for the Min-Max-Regret approach.

3.1 Optimality Criteria

Let π denote the true expected profit function, Π_0 — the set of all plausible profit functions, and $\mathcal{R}_0$ — the corresponding set of plausible reserve prices. The four prevalent approaches to modeling decisions under ambiguity are: Bayesian, Robust Bayesian, Max-Min, and Min-Max-Regret (Hirano and Porter, 2020; Manski, 2022).

A Bayesian decision-maker specifies a prior distribution Q over Π_0[10] and maxi-

[10] In our setting, it is natural to think of a prior distribution over the sharp identified set $\mathcal{F}_0$ for $F_{N:N}$, which implies a prior distribution over Π_0.

mizes the average profit,

$$\sup_{r\in\mathcal{R}_0}\int_{\Pi_0}\pi(r)dQ(\pi).$$

A Robust Bayesian decision-maker specifies a set $\mathcal{Q}$ of equally plausible prior distributions over Π_0 and maximizes the worst-case average profit,

$$\sup_{r\in\mathcal{R}_0}\inf_{Q\in\mathcal{Q}}\int_{\Pi_0}\pi(r)dQ(\pi).$$

Since Π_0 represents the sharp identified set for π, any prior distribution over Π_0 cannot be updated based on the data. Using different priors, one can recover any point $r \in \mathcal{R}_0$ as an optimal solution. Unless the seller has additional information that justifies the choice of a prior, we do not advocate for the Bayesian approaches.

A Max-Min (MM) decision-maker solves

$$\sup_{r\in\mathcal{R}_0}\inf_{\pi\in\Pi_0}\pi(r),$$

i.e., maximizes a lower bound on the expected profit. With i.i.d. values, assuming only that the transaction price in each auction is below the highest value, Aryal and Kim (2013) show that the Robust Bayes and MM criteria are equivalent. In some settings, the MM criterion may be interpreted as playing cautiously. However, this criterion does not utilize all of the available identifying information. For example, if the profit lower bound is monotonically decreasing, the MM reserve, $r = v_0$, is clearly suboptimal (Aradillas-López, Gandhi, and Quint, 2013). Even if additional restrictions substantially shrink the identified set but keep the lower bound fixed, the MM reserve stays the same. Empirically calibrated simulations suggest that the MM reserves are typically lower than the optimal ones (see also Jun and Pinkse, 2024). Nevertheless, we find that when applied with the new tight bounds on the expected profit, the MM solution performs well in practice.

A Min-Max-Regret (MMR) decision-maker solves

$$\inf_{r\in\mathcal{R}_0}\sup_{\pi\in\Pi_0}\{\pi(r^*_\pi)-\pi(r)\},$$

where $r^*_\pi \in \operatorname{argmax}_{r\in\mathcal{R}_0}\pi(r)$. Letting $\phi^*, \phi_r : \Pi_0 \to \mathbf{R}$ be defined as $\phi^*(\pi) = \pi(r^*_\pi)$,

$\phi_r(\pi) = \pi(r)$, the MMR problem can also be stated as

$$\inf_{r \in \mathcal{R}_0} \sup_{\pi \in \Pi_0} |\phi^*(\pi) - \phi_r(\pi)|. \tag{11}$$

That is, choosing the reserve price r is equivalent to choosing a functional $\phi_r(\cdot)$ as close as possible to the profit-maximizing functional $\phi^*(\cdot)$ uniformly over the set of unknown profit functions $\pi \in \Pi_0$. To the best of our knowledge, no MMR reserve prices are available in the existing literature. The MMR problem above is formulated with the sharp identified sets Π_0 and $\mathcal{R}_0$. When these are intractable, one may consider a "relaxed" problem with some outer sets Π and $\mathcal{R}$:

$$\inf_{r \in \mathcal{R}} \sup_{\pi \in \Pi} |\phi^*(\pi) - \phi_r(\pi)|. \tag{12}$$

The interpretation remains the same: one is choosing a functional $\phi_r(\cdot)$ as close as possible to the profit-maximizing functional $\phi^*(\cdot)$ uniformly over the set of unknown profit functions $\pi \in \Pi$. Intuitively, if $\Pi \backslash \Pi_0$ is not too large, and $\phi^*(\cdot)$ is well-behaved on $\Pi \backslash \Pi_0$, the solutions of (11) and (12) should be close to each other. Therefore, we view the relaxed MMR criterion in (12) as a reasonable way of choosing the reserve price under ambiguity.

3.2 The Min-Max-Regret Problem

In Theorem 1, we obtained bounds on $F_{N:N}$ under Assumptions 2.3 and 2.4. While the bounds are pointwise sharp, certain CDFs between the bounds are not admissible under the conditional independence assumption. Unfortunately, the sharp identified set for $F_{N:N}$ and thus for the expected profit function is hard to characterize, which makes the MMR problem in (11) intractable. To this end, we consider two relaxations of the MMR problem that can be solved analytically. Our results take bounds on $F_{N:N}$ as given, and hold for any bounds satisfying the stated assumptions, going beyond the setting of Theorem 1. The discussion is presented conditional on N, but the results extend to the unknown N case with obvious modifications.

3.2.1 Relaxed Min-Max-Regret Solution

Let $\pi_N(r; F_{N-1:N}, F_{N:N})$ denote the expected profit, conditional on N, defined in (2).

Assumption 3.1. *Let* $\mathcal{F} = \{F_{N:N} : \underline{F}(v) \leq F_{N:N}(v) \leq \overline{F}(v), \forall v \in \mathcal{V}\}$, *for some known nondecreasing functions* $\underline{F}(v)$, $\overline{F}(v)$ *satisfying* $0 \leq \underline{F}(v) \leq \overline{F}(v) \leq F_{N-1:N}(v)$, *for* $v \in \mathcal{V}$. *The set of admissible profit functions is* $\Pi = \{\pi_N(\,\cdot\,; F_{N-1:N}; F) : F \in \mathcal{F}\}$.

That is, a profit function π_N is admissible if it corresponds to some $F_{N:N}$ satisfying the available bounds. Plugging in $\overline{F}(r)$ and $\underline{F}(r)$ instead of $F_{N:N}(r)$ in Equation (2) yields the respective lower and upper bounds on π_N, but not all profit functions between the bounds are admissible. For example, π_N cannot arbitrarily "jump up" because $F_{N:N}(r)$ must be nondecreasing. Yet, the relaxed MMR solution is simple.

Theorem 5. *Let Assumptions 2.1, 2.2, and 3.1 hold. Then, for any* $r \geq v_0$, *the maximum regret function is given by* $R(r) = \max\{R_1(r), R_2(r)\}$, *where:*

$$R_1(r) = \max_{v_0 \leq s \leq r} \pi_N(s; F_{N-1:N}, F^*_{1,r}) - \pi_N(r; F_{N-1:N}, F^*_{1,r})$$

$$R_2(r) = \max_{s \geq r} \pi_N(s; F_{N-1:N}, F^*_{2,r}) - \pi_N(r; F_{N-1:N}, F^*_{2,r}),$$

with $F^*_{1,r}(v)$ *and* $F^*_{2,r}(v)$ *given by:*

$$F^*_{1,r}(v) = \mathbf{1}(v < r)\underline{F}(v) + \mathbf{1}(v \geq r)\overline{F}(v);$$

$$F^*_{2,r}(v) = \mathbf{1}(v < r)\overline{F}(v) + \mathbf{1}\{r \leq v < \bar{v}(r)\}\overline{F}(r) + \mathbf{1}\{v \geq \bar{v}(r)\}\underline{F}(v),$$

where $\bar{v}(r) = \min\{v : \underline{F}(v) \geq \overline{F}(r)\}$. *The MMR reserve is* $r^* = \operatorname{argmin}_{r \geq v_0} R(r)$.

The proof of this result is constructive. Given any reserve r, the maximum regret can be attained either below or above it. We show that the maximum regret below r is attained by a profit function that "jumps down" at r, corresponding to $F^*_{1,r}$, while the maximum regret above r is attained by a profit function that increases "as much as possible" above r, corresponding to $F^*_{2,r}$. Solving numerically for the MMR-optimal reserve r^* is straightforward. Typically, $R_1(r)$ is increasing and $R_2(r)$ is decreasing, at least locally around their intersection, so r^* solves $\Delta(r) = R_1(r) - R_2(r) = 0$.

Notably, as we show in Proposition 3 in Appendix, the relaxed MMR and max-min solutions often coincide.

3.2.2 Convex Min-Max-Regret Solution

Next, we impose an additional shape restriction on the set of admissible distributions of bidders value. Let $f_{V_{N:N}}(v)$ and $f_{V_{N-1:N}}(v)$ denote the density functions of the highest and second-highest values.

Assumption 3.2. *Suppose that the likelihood ratio $f_{V_{N:N}}(v)/f_{V_{N-1:N}}(v)$ is increasing in v. Let $\mathcal{F} = \{F = g(F_{N-1:N}) : g : [0,1] \to [0,1]$ is convex, $g(u) \in [\underline{\psi}(u), \overline{\psi}(u)]\}$, where $\underline{\psi}, \overline{\psi} : [0,1] \to [0,1]$ are increasing, twice continuously differentiable, and convex. The set of admissible profit functions is $\Pi = \{\pi_N(\,\cdot\,; F_{N-1:N}, F) : F \in \mathcal{F}\}$.*

The first part of Assumption 3.2 requires that $V_{N:N}$ dominate $V_{N-1:N}$ in the likelihood-ratio order. This restriction has a simple interpretation. Suppose an observer sees a realization $V = v$ drawn either from the distribution of $V_{N:N}$ or from that of $V_{N-1:N}$, but does not know which. Regardless of the observer's prior beliefs, a higher realization v increases the posterior odds that the observation was drawn from the distribution of the highest value. Thus, relatively high realizations are more indicative of $V_{N:N}$ than of $V_{N-1:N}$. The likelihood-ratio restriction constrains the shape of the joint distribution of bidders' values, but does not rule out either extreme of dependence. In particular, it holds under both independent private values (IPV) and perfectly correlated values (PCV), as well as for many commonly used dependence structures, including Gaussian, Clayton, and Frank copulas.[11] The second part of the assumption requires that the available bounds on $F_{N:N}$ are convex functions of $F_{N-1:N}$. It can be verified numerically for the bounds obtained in Theorem 1.

As we show in Lemma A.2 in Appendix, exchangeability of the valuations and likelihood ratio dominance imply that $F_{N:N}(v) = g_N(F_{N-1:N}(v))$, for some convex function $g_N : [0,1] \to [0,1]$, for all $v \in \mathcal{V}$. This fact motivates the stated definitions of $\mathcal{F}$ and Π in Assumption 3.2 and allows us to solve the MMR problem analytically.

Theorem 6. *Let Assumptions 2.1, 2.2, and 3.2 hold. Then, for any $r \geq v_0$, the*

[11] See, e.g., Nelsen (2006) or Joe (2014) for reviews of copula theory. The likelihood-ratio dominance is equivalent to $[V_{N:N} \,|\, a \leq V_{N:N} \leq b] \geq_{\text{st}} [V_{N-1:N} \,|\, a \leq V_{N-1:N} \leq b]$, for any $a, b \in \mathcal{V}$ with $a < b$, where $\geq_{\text{st}}$ denotes first-order stochastic dominance; see, e.g., Chapter 1.C. in Shaked and Shanthikumar (2007).

maximum regret is given by $R(r) = \max\{R_1(r), R_2(r)\}$, *with*

$$R_1(r) = \max_{v_0 \le v \le r} \pi_N(v; F_{N-1:N}, F^*_{1,r}) - \pi_N(r; F_{N-1:N}, F^*_{1,r})$$

$$R_2(r) = \max_{v \ge r} \pi_N(v; F_{N-1:N}, F^*_{2,r}) - \pi_N(r; F_{N-1:N}, F^*_{2,r}),$$

where $F^*_{j,r}(v) = g_{j,r}(F_{N-1:N}(v))$ *for convex functions* $g_{j,r} : [0,1] \to [0,1]$, $j \in \{1,2\}$, *defined as follows. Denoting* $u_r = F_{N-1:N}(r)$,

$$g_{1,r}(u) = \begin{cases} \underline{\psi}(u) & u \le \underline{u}_{1,r} \\ a_{1,r}u + b_{1,r} & \underline{u}_{1,r} < u \le u_r \\ \overline{\psi}(u) & u > u_r, \end{cases}$$

where $\underline{u}_{1,r}$ *solves* $\overline{\psi}(u_r) = \overline{\psi}'(u_r)(u_r - \underline{u}_{1,r}) + \underline{\psi}(\underline{u}_{1,r})$, $a_{1,r} = \overline{\psi}'(u_r)$, *and* $b_{1,r} = \overline{\psi}(u_r) - a_{1,r}u_r$. *Further:*

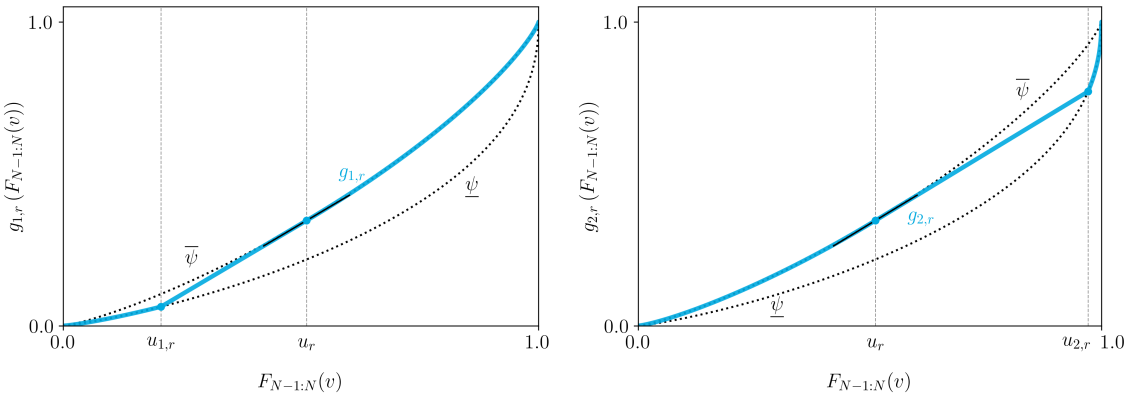
$$g_{2,r}(u) = \begin{cases} \overline{\psi}(u) & u \le u_r \\ a_{2,r}u + b_{2,r} & u_r < u \le \overline{u}_{2,r} \\ \underline{\psi}(u) & u > \overline{u}_{2,r}, \end{cases}$$

where $\bar{u}_{2,r}$ *solves* $\underline{\psi}(\bar{u}_{2,r}) - \overline{\psi}'(u_r)(\bar{u}_{2,r} - u_r) = \overline{\psi}(u_r)$, $a_{2,r} = \overline{\psi}'(u_r)$, *and* $b_{2,r} = \overline{\psi}(u_r) - a_{2,r}u_r$. *The Min-Max-Regret reserve is* $r^* = \operatorname{argmin}_{r \ge v_0} R(r)$.

The proof idea is similar to that of Theorem 5, except the "worst-case" CDFs $F^*_{1,r}$ and $F^*_{2,r}$ are now required to be convex transformations of $F_{N-1:N}$. The transformation $g_{1,r}$ corresponds to the CDF that decreases "as fast as possible" just below r, and $g_{2,r}$ to the CDF that increases "as fast as possible" above r, while preserving convexity. Figure 1 illustrates. As in Theorem 5, solving numerically for r^* is straightforward.

4 Estimation and Inference

For brevity, the exposition below treats the number of bidders N as known and fixed, but the extensions to varying or unknown N are immediate. We assume that the econometrician has access to a sample of n independent auctions, $(S_i, X_i)_{i=1}^n$, where $X_i \in \mathcal{X}$ is vector of auction-specific covariates, and $S_i \in \mathbf{R}_+$ is the transaction price in auction i. Further, we assume that the bounding functions $\underline{D}_N(v)$ and $\overline{D}_N(v)$ are



(a) Function $g_{1,r}$ defining $R_1(r)$ (b) Function $g_{2,r}$ defining $R_2(r)$

Figure 1: Bounding functions in Theorem 6

known. We defer all formal statements and proofs to Appendix D and only provide intuition and practical recommendations below.

4.1 Bounds on Expected Profit

The profit bounds in Theorem 2 are known, deterministic functions of the observable distribution $F_{N-1:N}(v) = P(V_{N-1:N} \leq v \,|\, N)$. The best approach to estimating this distribution depends on the context. In addition to the number of bidders N, one may want to condition on a vector of covariates X. If the sample size is limited, or X is very rich, nonparametric estimation may be imprecise. In such cases, one may impose a flexible parametric model for $P(V_{N-1:N} \leq v \,|\, N, X)$, similar to Athey, Levin, and Seira (2011), or assume that $P(V_{N-1:N} \leq v \,|\, N, X)$ depends on X only through an index $X'\gamma$ and employ, e.g., the semiparametric MLE of Ai (1997). When nonparametric estimation is suitable and some covariates are continuous, the bandwidths may be selected using a reference parametric model; see Ichimura and Todd (2007).

Letting $\underline{\pi}_N(r; F_{N-1:N})$ and $\overline{\pi}_N(r, F_{N-1:N})$ denote the profit bounds in Theorem 2 and $\hat{F}_{N-1:N}$ be a suitable estimator for $F_{N-1:N}$, the profit bounds are estimated as:

$$\hat{\underline{\pi}}_N(r) = \underline{\pi}_N(r; \hat{F}_{N-1:N}); \qquad \hat{\overline{\pi}}_N(r) = \overline{\pi}_N(r; \hat{F}_{N-1:N}). \tag{13}$$

Since the bounding functions $\underline{\psi}_N$ and $\overline{\psi}_N$ are convex, the plug-in estimators of the profit bounds are generally biased downward in small samples. The bias may be corrected using jackknife or bootstrap techniques; see, e.g., Wasserman (2006). pointwise

and uniform confidence bands can be constructed using the delta method.

Our empirical application features only discrete covariates, so we use the sample analog estimator $\hat{F}_{N-1:N}$ within each covariate bin. In simulations with comparable sample sizes, we find that the finite-sample bias of the plug-in estimates $\underline{\psi}_N(\hat{F}_{N-1:N})$ and $\overline{\psi}_N(\hat{F}_{N-1:N})$ is negligible, and thus do not apply bias correction. Appendix D spells out the asymptotic distributions of $\hat{\underline{\pi}}_N(r)$ and $\hat{\overline{\pi}}_N(r)$ in this setting.

4.2 Expected Profit Under MMR Reserve Price

First, consider estimating the MMR optimal reserve price r^* obtained in Theorems 5 or 6. Letting $R_j(r; F_{N-1:N})$ for $j \in \{1, 2\}$ denote the maximum regret functions, the MMR optimal reserve price r^* solves:[12]

$$\Delta(r; F_{N-1:N}) \equiv R_1(r; F_{N-1:N}) - R_2(r; F_{N-1:N}) = 0.$$

Denoting $\hat{\Delta}(r) = \Delta(r; \hat{F}_{N-1:N})$, a natural estimator $\hat{r}$ for r^* solves:

$$\hat{\Delta}(\hat{r}) = 0. \tag{14}$$

Under the standard regularity conditions, consistency of $\hat{r}$ follows from the uniform consistency of $\hat{F}_{N-1:N}$ and continuity of the mapping $F \mapsto \Delta(\,\cdot\,; F)$ with respect to the sup-norms. In Appendix D, under further regularity conditions, using the functional delta method, we further establish that

$$\sqrt{n}(\hat{r} - r^*) \quad \to_d \quad N\left(0, V_r\right). \tag{15}$$

Next, consider the expected profit under MMR reserve price, $\pi^* = \pi_N(r^*)$. It is bounded by $\pi^* \in [\underline{\pi}^*, \overline{\pi}^*] \equiv [\underline{\pi}_N(r^*), \overline{\pi}_N(r^*)]$. The natural estimators for the profit bounds are:

$$\hat{\underline{\pi}}^* = \hat{\underline{\pi}}(\hat{r}); \qquad \hat{\overline{\pi}}^* = \hat{\overline{\pi}}(\hat{r}), \tag{16}$$

where $\hat{\underline{\pi}}$ and $\hat{\overline{\pi}}$ are defined in (13), and $\hat{r}$ in (14). Consistency of these estimators follows from consistency of $\hat{r}$ and continuity of the maps $F \mapsto \underline{\pi}(\cdot; F)$ and $F \mapsto \overline{\pi}(\cdot; F)$ with respect to the sup-norms. For the asymptotic distribution, consider an

[12] That is, the minimizer of $R(r) = \max(R_1(r), R_2(r))$ is given by the intersection of $R_1(r)$ and $R_2(r)$.

expansion:

$$\begin{aligned}\sqrt{n}(\hat{\underline{\pi}}^* - \underline{\pi}^*) &= \{\sqrt{n}(\hat{\underline{\pi}}(\hat{r}) - \underline{\pi}(\hat{r})) - \sqrt{n}(\hat{\underline{\pi}}(r^*) - \underline{\pi}(r^*))\} \quad (I) \\ &+ \sqrt{n}(\hat{\underline{\pi}}(r^*) - \underline{\pi}(r^*)) \quad (II) \\ &+ \sqrt{n}(\underline{\pi}(\hat{r}) - \underline{\pi}(r^*)). \quad (III)\end{aligned} \tag{17}$$

Using stochastic equicontinuity, term (I) can be shown to converge in probability to zero. Following the preceding discussion, $(II) + (III)$ is asymptotically normal. Applying similar arguments to the upper bound estimator $\hat{\overline{\pi}}^*$, it follows that:

$$\begin{aligned}\sqrt{n}(\hat{\underline{\pi}}^* - \underline{\pi}^*) &\to_d \mathcal{N}(0, V_{\underline{\pi}^*}); \\ \sqrt{n}(\hat{\overline{\pi}}^* - \overline{\pi}^*) &\to_d \mathcal{N}(0, V_{\overline{\pi}^*}).\end{aligned} \tag{18}$$

Thus, given consistent estimators $\hat{V}_{\underline{\pi}}$ and $\hat{V}_{\overline{\pi}}$, a $(1-\alpha)$ confidence interval for $\pi^* \in [\underline{\pi}^*, \overline{\pi}^*]$ can be constructed as: $[\hat{\underline{\pi}}^* - z_{1-\alpha}\hat{V}_{\underline{\pi}^*}^{1/2}/n^{1/2}, \hat{\overline{\pi}}^* + z_{1-\alpha}\hat{V}_{\overline{\pi}^*}^{1/2}/n^{1/2}]$. Such a confidence interval is uniformly asymptotically valid over a large set of distributions, away from point identification; See Imbens and Manski (2004), Stoye (2009), and Canay and Shaikh (2017) for related discussions. In our setup, point identification is ruled out by construction of the bounds, so the above confidence interval is appropriate.

The asymptotic variances in (18) are cumbersome due to the dependence between terms (II) and (III) in (17). However, we show that the finite-sample distributions of $\sqrt{n}(\hat{\underline{\pi}}^* - \underline{\pi}^*)$ and $\sqrt{n}(\hat{\overline{\pi}}^* - \overline{\pi}^*)$ change smoothly with the underlying distribution of the data, so a valid confidence interval can be constructed using the bootstrap.

Algorithm 1.

1. *Using the original sample* $(S_i, X_i)_{i=1}^n$, *compute* $\hat{\underline{\pi}}_N^*$ *and* $\hat{\overline{\pi}}_N^*$ *according to* (16).

2. *Draw a large number of bootstrap samples* $(S_{i,b}^*, X_{i,b}^*)_{i=1}^n$, *for* $b \in \{1, \ldots, B\}$, *and compute* $\hat{\underline{\pi}}_b^*$ *and* $\hat{\overline{\pi}}_b^*$ *according to* (16), *for each* b.

3. *Let* $\hat{\underline{q}}_{1-\alpha}$ *denote the* $(1-\alpha)$ *empirical quantile of* $\{\sqrt{n}(\hat{\underline{\pi}}_b^* - \hat{\underline{\pi}}^*)\}_{b=1}^B$, *and* $\hat{\overline{q}}_\alpha$ *denote the* α *empirical quantile of* $\{\sqrt{n}(\hat{\overline{\pi}}_b^* - \hat{\overline{\pi}}^*)\}_{b=1}^B$, *conditional on the data. Compute the confidence interval:*

$$\left[\hat{\underline{\pi}}^* - \frac{\hat{\underline{q}}_{1-\alpha}}{\sqrt{n}}, \ \hat{\overline{\pi}}^* - \frac{\hat{\overline{q}}_\alpha}{\sqrt{n}}\right]. \tag{19}$$

■

We remark that although Step 2 involves solving the MMR problem for each bootstrap sample, the overall procedure is computationally simple.

5 Monte Carlo Experiments

We illustrate the performance of the proposed approach in a series of simulations. We show that the CDF bounds in Theorems 1 and 3 and the implied profit bounds in Theorems 2 and 4 are substantially tighter than those existing in the literature. Further, we show that choosing the reserve price using either max-min or minimax-regret criteria in Theorems 5 and 6, when applied with the new tighter bounds, may lead to a substantial increase in seller's profit.

5.1 Simulation Design

We calibrate the simulation design to four stylized facts in our empirical setting.

First, the auction houses report the low and the high estimates of each lot's value. These estimates are crude and not necessarily backed by any market or historical research. With the high estimate normalized to 1, the low estimate is around 2/3, on average. We set the value of unsold good to the seller to the midpoint between low and high estimates, $v_0 = 5/6$.

Second, the reserve price is secret and does not exceed the lot's low estimate. In simulations, we set the reserve price to zero.

Third, there is significant variation in the number of active bidders per auction. We randomly draw the number of bidders N for each auction from the set $N \in \{2, 3, \ldots, 10\}$, and use $\underline{N} = 2$ and $\overline{N} = 10$ throughout the simulations.

Fourth, according to the bidding guidelines, the minimal bid increment is 10% of the standing bid, but this rule is not strictly enforced. We simulate auctions as follows. Given the initial set of N bidders, the first bid by a randomly drawn bidder is cast at 5% of their value. At each subsequent iteration, an active bidder is drawn at random from the pool of remaining bidders, whose values are above the standing bid. With probability $1 - \lambda$, the new bid is equal to 1.1 of the standing bid, and with probability λ, the new bid is drawn uniformly between the standing bid and 1.1 of the standing bid. If the new bid is above the active bidder's value, the bidder

is eliminated, and a new bidder is drawn from the pool of remaining bidders. The iterations proceed until only one bidder remains.

5.2 Simulation Results

First, we illustrate the CDF and profit bounds from Theorems 3 and 4. The valuations are generated as $\log(V_j) = U + \varepsilon_j$, where $U \sim N(\mu_U, \sigma_U^2)$ and the distribution of ε_j is such that $\varepsilon_{N-1:N} \sim N(\mu_\varepsilon, \sigma_\varepsilon^2)$. The parameters $\mu_U, \mu_\varepsilon, \sigma_U$, and σ_ε are chosen to match the average transaction price in the data, $\mathbb{E}[V_{N-1:N}] = 1.35$. The copula function used to define the bounds in Assumption 2.4 matches the above data-generating process and corresponds to $\rho = 0.8$. We use a kernel-smoothed estimator for the CDF of the sale price with bandwidth $h_n = n^{-1/3}$ (see Appendix D for the details).

Figure 2 presents the unconditional bounds on the CDF of $V_{N:N}$ and the corresponding bounds on expected profit. The black solid line represents the true CDF, and the black dashed lines the IPV and PCV bounds. The blue dashed lines represent lower and upper bounds with $\underline{\rho} = \overline{\rho} = 0.8$, and the green lines with $\underline{\rho} = 0.7$ and $\overline{\rho} = 0.9$. The proposed bounds are substantially tighter than the IPV and PCV bounds even for a relatively wide range of dependence, $[\underline{\rho}, \overline{\rho}] = [0.7, 0.9]$. On the other hand, even when correctly specifying $\underline{\rho} = \overline{\rho} = 0.8$, the bounds are still far from point identification. The results conditional on N are very similar and thus omitted.

Next, we compare the MMR reserves of Theorems 5 and 6 with the MM reserves and the *status quo* of $r = v_0$. We consider $\underline{\rho} = 0, \overline{\rho} = 1$ and $\underline{\rho} = \overline{\rho} = 0.8$, and $v_0 = 5/6$. Figure 3 and Table 1 present the results. The left panel depicts the MM reserves, which maximize the corresponding profit lower bounds. With $\underline{\rho} = 0, \overline{\rho} = 1$, the lower bound is monotonically decreasing, so the MM reserve is v_0. With $\underline{\rho} = \overline{\rho} = 0.8$, the MM reserve is 1.02, which appears reasonable and is much closer to the true optimum of 1.27. The right panel depicts the maximum regret functions from Theorems 5 and 6 and corresponding MMR reserves. With $\underline{\rho} = 0, \overline{\rho} = 1$, we find that both Relaxed MMR and Convex MMR reserves coincide with v_0. In contrast, with $\underline{\rho} = \overline{\rho} = 0.8$, the relaxed MMR reserve coincides with MM (1.02), while the convex MMR reserve is much closer to the optimum (1.22 vs. 1.27).

Thus, we find that the MM, relaxed MMR, and especially Convex MMR reserves computed with the new tight bounds on the CDF of the highest value are close to the optimum and result in a sizable increase in expected profit.

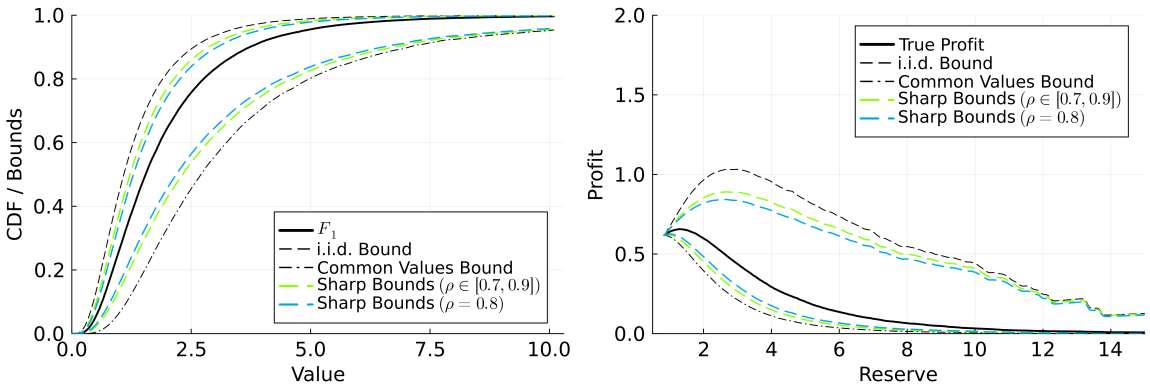


Figure 2: Simulated bounds on the CDF of the highest value and expected profit.

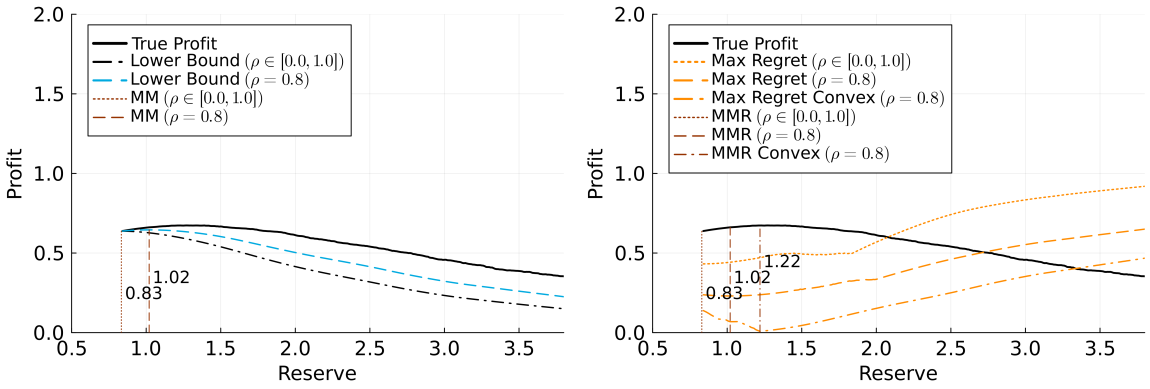


Figure 3: Simulated profit bounds and optimal reserve prices.

	Reserve	**Profit**	**Increase over $r = v_0$**
True Optimal	1.27	0.673	5.49%
MM ($\underline{\rho} = 0.8, \overline{\rho} = 0.8$)	1.02	0.662	3.63%
MM/Both MMR ($\underline{\rho} = 0.0, \overline{\rho} = 1.0$)	0.83	0.638	0.00%
Relaxed MMR ($\underline{\rho} = 0.8, \overline{\rho} = 0.8$)	1.02	0.662	3.63%
Convex MMR ($\underline{\rho} = 0.7, \overline{\rho} = 0.9$)	0.95	0.654	2.43%
Convex MMR ($\underline{\rho} = 0.8, \overline{\rho} = 0.8$)	1.22	0.673	5.42%

Table 1: Simulated profit under suggested reserve prices.

6 Optimal Reserve Prices in Art Auctions

6.1 Data

We construct a new dataset of auctions held by Christie's and Sotheby's, the two largest auction houses selling art and antiquities. The auctions follow an open ascending format, with thirty lots presented in each auction, on average. We treat each lot as a separate auction. On their websites, the auction houses publish a detailed description of each lot, including its dimensions, medium, whether it is signed by the author, ownership and auction history, and references to existing literature and past exhibitions. In addition, the auction houses publish "low" and "high" estimates of each lot's value. The websites indicate that these estimates are not necessarily backed by market research and only serve as a crude reference for prospective bidders.

The reserve price is kept secret but is known to be less than the "low" estimate of the lot's value. To ensure that the reserve is met, the auctioneer may bid on behalf of the seller. For some lots, the auction houses offer a minimum price guarantee to the seller, regardless of the final sale. In such cases, to share the risks of underselling, the auction houses may invite a third party to submit an irrevocable bid.

The auction houses provide extensive information about individual lots but relatively little about the auction characteristics and the bidding process. Although the location and date of the auction and final sale prices are typically published on the websites, historical data on bids, the number of bidders, and bidder identities remain private. To overcome this limitation, we use publicly available livestream recordings on YouTube, which cover more than 3,500 lots sold between 2020 and 2024.

6.1.1 Bids and Lot Information

For each auction lot, we obtain a complete bidding trajectory by applying computer vision techniques to frame-by-frame video data. Figure 4 presents examples of the frames. We crop the bottom sections, highlighted in red, and use an optical character recognition package Tesseract[13] to extract the lot's information and the bidding sequence. Then, we match the bids for each lot with the lot's characteristics scraped from the auction houses' websites. We use the Ratcliff-Obershelp strings matching algorithm[14] to accommodate any errors or misspellings in Tesseract's output. Table

[13] Available at https://github.com/tesseract-ocr/tesseract.

[14] Available in the `SequenceMatcher` class in Python's `difflib` module

Category	Subcategory	Location	Count
Art	Chinese Art	New York	112
Art	Impressionist/20th/21st Century Art	Hong Kong	636
Art	Impressionist/20th/21st Century Art	Las Vegas	9
Art	Impressionist/20th/21st Century Art	London	708
Art	Impressionist/20th/21st Century Art	New York	1208
Art	Impressionist/20th/21st Century Art	Shanghai	55
Art	Old Masters	London	102
Art	Old Masters	New York	134
Wine/jewelry/etc.	–	–	542
Total			3506

Table 2: The art auction dataset.

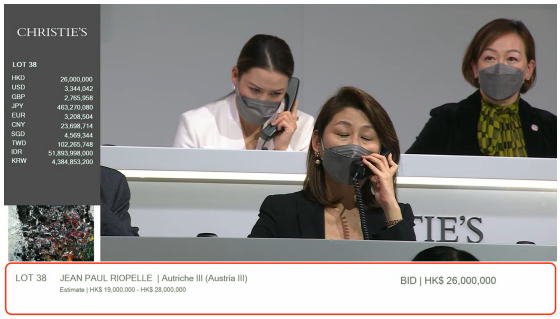
2 summarizes the auctions across art categories and locations. Due to the nature of the data, we cannot match the bids with the bidders. Thus, we only use the two highest bids, which we assume belong to different bidders.

6.1.2 Number of Bidders

The exact number of bidders in each auction cannot be reliably identified from the auction videos, so we obtain bounds on it using the audio transcripts.[15] There are four types of bidders in each auction: (i) telephone, (ii) live, (iii) absentee, and (iv) online; see Figure 5. We obtain a lower bound the true number of bidders by bounding the number of unique bidders of each type.

Each telephone bidder is represented by an employee of the auction house, whom the auctioneer calls by their first names. For example, in the top-right panel of Figure 5, the auctioneer references a Christie's employee, Olivier Camu, putting a telephone bid on behalf of an unknown buyer. To identify the names from the transcript, we use the Named Entity Recognition language models. We apply two state-of-the-art language models, RoBERTa (Liu, Ott, Goyal, Du, Joshi, Chen, Levy, Lewis, Zettlemoyer, and Stoyanov, 2019) and Contextual String Embeddings (Akbik, Blythe, and Vollgraf, 2018), and combine their output matching similar strings and removing duplicates. To capture the identities of the unique live, absentee, and online bidders, we look for keywords {"online bidder" = "online," "sir" = "gentleman," "madam" =

[15]The transcripts are generated using Google Cloud Platform Speech-to-Text Recognition model. See https://cloud.google.com/speech-to-text for the details.



Lot 38, Christie's Hong Kong, Nov 2022



Lot 117, Sotheby's New York, Nov 2022

Figure 4: Example screenshots taken from Christie's and Sotheby's live auctions posted on YouTube. Red boxes highlight the relevant portions of the screen.

"lady," "back of the room," "to the right," "to the left," "absentee"}. This approach allows us to capture all of the telephone bidders, up to one unique online or absentee bidder, and up to five unique live bidders.

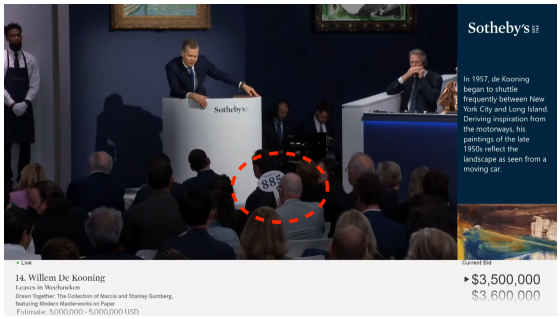
The resulting median lower bound on the number of bidders is five; see Table 3. The distribution of the lower bound is shown in Figure 6 in Appendix. Although it is unclear how tight the resulting lower bound is, the existing statistics are reassuring: in 2021, 42% of winning bids in Christie's live auctions were made over the telephone and only 7% in the salesroom, meaning that at least some of the "missing" bidders may not have affected the auction's outcome anyway.[16]

We conservatively set the upper bound on the number of bidders as two times the lower bound. An even more conservative option would be the total number of bids. However, we find that the function $\phi_N(t)^N$, which is used to construct the bounds, converges pointwise as N increases, so for $N, N' \geq 10$ the difference between $\phi_N(t)^N$ and $\phi_{N'}(t)^{N'}$ becomes negligible. Thus, the choice between upper bounds above ten bidders does not affect the results.

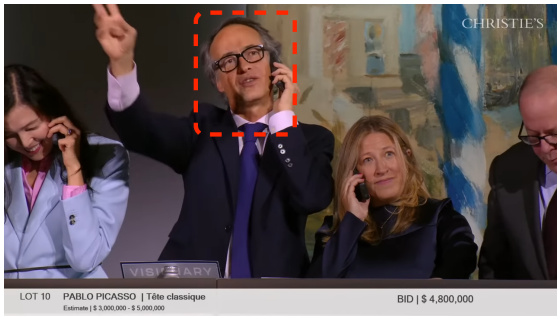
6.1.3 Auction House Fees

The auction houses charge around 25% buyer's premium and 10% seller's commission with some variation depending on location and lot type. Table 6 in Appendix summarizes the Buyer's Premium Schedule, for both Christie's and Sotheby's, as of March 2023. We adjust the bids accordingly.

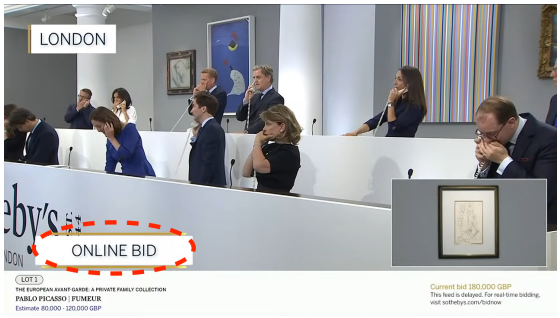
[16] See, e.g., https://www.nytimes.com/2018/10/02/arts/auctions-live-online-bidding.html.



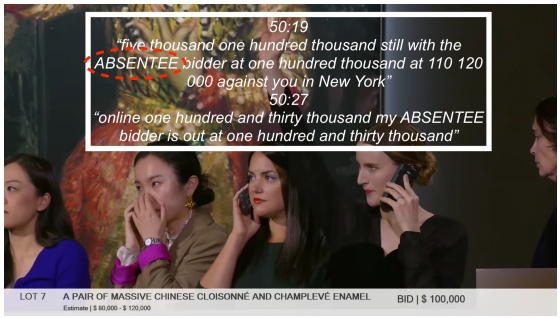
Room



Telephone



Online



Absentee

Figure 5: The four types of bidders in Sotheby's or Christie's live auctions.

6.2 Modern Art Sold in New York City

Our analysis focuses on modern art sold in New York City. This is the largest subsample in our data both in terms of the number of auctions and the generated revenue. We exclude outliers and focus on the lots with presale estimates between \$100K and \$10.0M, which leaves us with 949 observations. We further condition on the bin of the high presale estimate, $[\$100\text{K}, \$1.0\text{M}]$ or $[\$1.0\text{M}, \$10.0\text{M}]$, which roughly splits the sample into one-third and two-thirds, and the bin of number of bidders, $[2, 10]$, $[6, 40]$, or $[2, 40]$ (i.e., $\underline{N} \in [2, 5]$, $\underline{N} \in [6, 20]$, or $\underline{N} \in [2, 20]$), which further splits each of the subsamples approximately in half. Table 3 displays summary statistics. Notably, the median and average sale prices are both substantially larger than the "high" estimates of the lot's value, revealing that the presale estimates reported by the auction houses are conservative.

The assumptions required for our theoretical results are plausible in this empirical setting. We do not observe any systematic differences in bidder's behavior or identities, which could be exploited in the analysis. Moreover, prior to each auction, the bidders have an opportunity to view the objects being sold and assess their quality

Presale Estimates ∈ [$100K, $1.0M]; Number of lots: 287				
Variable	**Median**	**Mean**	**Std**	**[Min, Max]**
Transaction price/High estimate	1.39	1.83	1.29	[0.25, 6.35]
Second-highest bid	1.31	1.72	1.23	[0.24, 6.14]
Number of bids	10.00	11.42	7.40	[2.00, 39.00]
Number of bidders	5.00	5.44	2.50	[2.00, 13.00]
Low estimate/High estimate	0.67	0.69	0.05	[0.60, 0.78]
Presale Estimates ∈ [$1.0M, $10.0M]; Number of lots: 662				
Variable	**Median**	**Mean**	**Std**	**[Min, Max]**
Transaction price/High Estimate	1.15	1.41	0.85	[0.37, 6.72]
Second-highest bid	1.11	1.34	0.82	[0.35, 6.30]
Number of bids	10.00	11.75	8.04	[2.00, 76.00]
Number of bidders	5.00	5.98	2.91	[2.00, 20.00]
Low estimate/High estimate	0.67	0.68	0.05	[0.48, 0.84]

Table 3: Summary statistics for modern art auction lots in New York City.

personally, or employ a consultant. As a result, assuming symmetric private values (Assumption 2.1) appears reasonable. Further, examining the bidding data, we find that (i) There is no evidence of jump bids towards the end of the auctions; (ii) The difference between the two highest bids is small, relative to the size of the bids; (iii) The auctioneers do not enforce the minimal bid increment guideline towards the end of the auction; and (iv) All lots in the retained sample are sold. Therefore, the assumption that the transaction price reveals second-highest valuation (Assumption 2.2) also appears plausible.

While we control for the art category, location, presale estimates range, and bounds on the number of bidders, the bidders also take into account detailed information about the lots when forming their private values. The difference between the information we use in the analysis and the information available to the bidders contributes to the unobserved auction-level heterogeneity (Assumption 2.3). Since we observe at least part of the detailed lot information, we can empirically calibrate the magnitude of the unobserved heterogeneity, measured by the parameters $\underline{\rho}, \overline{\rho}$ (Assumption 2.4). Specifically, we calibrate $\underline{\rho}$ by comparing out-of-sample performance of two prediction models for the sale price, one with the *short* list of covariates we

actually use in estimation, and one with the *long* list of all of the covariates available to the bidders. The details are provided in Appendix B. As we show in Lemma B.1, under further mild assumptions, the quantity

$$\underline{\rho} = 1 - \frac{\text{MSE}_{\text{long}}}{\text{MSE}_{\text{short}}} \tag{20}$$

provides a valid lower bound on the parameter ρ in Assumption 2.4. Intuitively, if MSE_{long} is much smaller than $\text{MSE}_{\text{short}}$, the analyst leaves a lot of information on the table, corresponding to larger $\underline{\rho}$. Depending on the subsample, the calibrated values of $\underline{\rho}$ are between 0.4 and 0.7, and we consider $\overline{\rho} \in \{\underline{\rho}, \underline{\rho} + 0.1\}$.

Table 4 presents our empirical results. All non-dollar amounts reported in the table are normalized by the average high estimate for the corresponding subsample. We set the value of unsold good to the seller at $v_0 = 5/6$, corresponding to the midpoint between the low and high estimates. Since the median transaction prices for all subsamples is well above one (see Table 3), this choice can be viewed as conservative.[17]

The bottom part of Table 4 reports the results with unrestricted dependence between the values, $[\underline{\rho}, \overline{\rho}] = [0, 1]$. In this case, both the Convex MMR and max-min reserves are equal to v_0. This fact remains true for other values of v_0 (see, e.g., Table 7 in Appendix), suggesting that the unrestricted bounds are not sufficiently informative to guide the choice of the reserve price.

The top part of Table 4 reports the results assuming that $\rho = \overline{\rho} = \underline{\rho}$, with $\underline{\rho}$ calibrated in (20). We find that the convex MMR reserves are consistently higher than the max-min reserves, and both reserves are higher than the high estimate (which is one, due to normalization). The estimated profit bounds under the max-min reserve are included within the profit bounds under the Convex MMR reserve. For both reserves, the estimated lower bounds on expected profit are consistently higher than the average realized profit, with the projected profit gains being larger for higher-priced lots with fewer bidders.

Finally, the middle part of Table 4 reports the results for $\rho \in [\underline{\rho}, \underline{\rho} + 0.1]$. Compared with the top part of the table, both reserves and the corresponding bounds on expected profit are lower, but all qualitative comparisons remain the same. The projected profit gains are the largest for the lots within the [1\$M, 10\$M] presale high

[17] Table 7 in Appendix additionally reports the results for v_0 equal to the average low estimate.

estimate range and $N \in [2, 10]$ bidders. With the Convex MMR approach, we find that setting the reserve price 37% above the projected value of unsold lot would increase the expected profit by approximately [\$150K, \$489K] or $[26.66\%, 86.66\%]$ of the realized average profit per lot. With the max-min approach, we find that setting the reserve price 12% above the projected value of unsold lot would increase the expected profit by approximately [\$225K, \$299K], or $[40.00\%, 53.00\%]$ of the realized average profit, per lot. For this subsample, the entire 95% confidence intervals for both types of reserves are above the realized profit.

7 Conclusion and Further Research

This paper proposed a novel approach to partial identification of the distribution of the highest valuation, expected profit, and consumer surplus in open ascending auctions. We derived new bounds on the distribution of the highest value and seller's profit, bounding the degree of dependence between the values, and proposed a way to select a reserve price under ambiguity using the minimax-regret criterion. The proposed approach is computationally simple and readily extends to settings with unknown number of bidders, as long as bounds on it are observed or can be conservatively set. We demonstrated that the resulting bounds are substantially tighter than those previously available in the literature, and the proposed method of selecting a reserve price performs well in simulations. We applied the methodology to a large new dataset of art auctions held by Christie's and Sotheby's and argued that higher reserve prices could lead to a sizable increase in profit.

Our analysis can be advanced in several directions. First, one may be interested in welfare aspects of the chosen auction format. Under the stated assumptions, bidders surplus is also identified from the marginal distributions of the two highest values and can be studied similarly to the expected profit. Second, the assumption that the transaction price reveals the second-highest value in the observed data can be relaxed to weaker bidding assumptions in the spirit of Haile and Tamer (2003). In such setting, bounds on the distribution of the highest value follow directly from the arguments above, but the analysis of optimal reserve prices becomes more nuanced. A coherent formulation of expected profit requires additional assumptions on bidding behavior in counterfactual auctions, and the minimax-regret problems for selecting the reserve price become more complex. Finally, an interesting direction for further

High Estimate Range	[$100K, $1.00M]			[$1.00M, $10.00M]		
Number of Bidders	$N \in$ [2,10]	$N \in$ [6,40]	$N \in$ [2,40]	$N \in$ [2,10]	$N \in$ [6,40]	$N \in$ [2,40]
Avg High Estimate	$532,443	$483,529	$507,771	$3,761,415	$3,641,264	$3,696,885
Avg Realized Profit	0.58	1.49	1.07	0.15	0.78	0.45
$[\underline{\rho}, \overline{\rho}]$	[0.70, 0.70]	[0.40, 0.40]	[0.50, 0.50]	[0.60, 0.60]	[0.65, 0.65]	[0.55, 0.55]
Convex MMR reserve	1.44	1.74	1.66	1.21	1.02	1.61
– 95% CI (reserve)	[1.00, 1.51]	[1.29, 1.85]	[1.50, 1.79]	[0.98, 1.25]	[1.00, 1.53]	[1.10, 1.67]
Bounds on profit	[0.57, 0.73]	[1.50, 1.57]	[1.07, 1.24]	[0.18, 0.29]	[0.80, 0.81]	[0.44, 0.66]
– 95% CI (profit)	[0.47, 0.85]	[1.33, 1.73]	[0.96, 1.35]	[0.16, 0.33]	[0.71, 0.91]	[0.40, 0.71]
Max-min reserve	0.99	1.34	1.17	0.95	1.02	1.04
– 95% CI (reserve)	[0.95, 1.03]	[1.24, 1.44]	[1.13, 1.22]	[0.93, 0.98]	[0.99, 1.05]	[1.01, 1.07]
Bounds on profit	[0.63, 0.65]	[1.52, 1.54]	[1.11, 1.15]	[0.21, 0.24]	[0.80, 0.81]	[0.50, 0.54]
– 95% CI (profit)	[0.52, 0.77]	[1.35, 1.70]	[1.01, 1.26]	[0.18, 0.27]	[0.71, 0.90]	[0.46, 0.59]
$[\underline{\rho}, \overline{\rho}]$	[0.70, 0.80]	[0.40, 0.50]	[0.50, 0.60]	[0.60, 0.70]	[0.65, 0.75]	[0.55, 0.65]
Convex MMR reserve	1.27	1.55	1.47	1.14	0.98	1.46
– 95% CI (reserve)	[0.94, 1.33]	[1.21, 1.67]	[1.33, 1.59]	[0.93, 1.17]	[0.96, 1.40]	[1.05, 1.51]
Bounds on profit	[0.59, 0.70]	[1.50, 1.56]	[1.08, 1.21]	[0.19, 0.28]	[0.80, 0.81]	[0.44, 0.63]
– 95% CI (profit)	[0.49, 0.83]	[1.34, 1.72]	[0.97, 1.31]	[0.16, 0.32]	[0.71, 0.91]	[0.40, 0.68]
Max-min reserve	0.94	1.25	1.10	0.93	0.98	0.99
– 95% CI (reserve)	[0.92, 0.97]	[1.18, 1.33]	[1.07, 1.14]	[0.91, 0.95]	[0.96, 1.00]	[0.97, 1.01]
Bounds on profit	[0.62, 0.64]	[1.52, 1.53]	[1.11, 1.14]	[0.21, 0.23]	[0.80, 0.81]	[0.50, 0.53]
– 95% CI (profit)	[0.51, 0.77]	[1.35, 1.70]	[1.00, 1.25]	[0.18, 0.27]	[0.72, 0.90]	[0.45, 0.58]
$[\underline{\rho}, \overline{\rho}]$	[0.00, 1.00]	[0.00, 1.00]	[0.00, 1.00]	[0.00, 1.00]	[0.00, 1.00]	[0.00, 1.00]
Reserve*	0.83	0.83	0.83	0.83	0.83	0.83
– 95% CI (reserve)	[0.83, 0.83]	[0.83, 0.83]	[0.83, 0.83]	[0.83, 0.83]	[0.83, 0.83]	[0.83, 0.84]
Bounds on profit	[0.62, 0.62]	[1.50, 1.50]	[1.10, 1.10]	[0.20, 0.20]	[0.80, 0.80]	[0.49, 0.49]
– 95% CI (profit)	[0.51, 0.75]	[1.34, 1.67]	[1.00, 1.22]	[0.17, 0.23]	[0.71, 0.88]	[0.45, 0.54]
Observations	131	153	314	311	269	610

*For the unrestricted range, $[\underline{\rho}, \overline{\rho}] = [0,1]$, the convex MMR and max-min reserves coincide.

Table 4: Convex minimax-regret and max-min reserves with profit bounds (and bootstrap 95% CIs) for Modern Art, New York. The average realized profit, bounds on expected profit, and 95% CIs are scaled by high estimate.

research is allowing for asymmetric bidders. While our bounding approach critically relies on the values being conditionally i.i.d., it appears possible to accommodate several "types" of symmetric bidders using similar ideas.

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A Proofs of the main results

We begin with an auxiliary lemma that is instrumental in the proofs of the main results.

Lemma A.1. *Suppose $N > 2$. Consider a function $h_N : [0,1] \to [0,1]$ given by:*

$$h_N(t) = \phi_N(t)^N,$$

where $\phi_N : [0,1] \to [0,1]$ is defined implicitly via $t = N\phi_N(t)^{N-1} - (N-1)\phi_N(t)^N$. Then:

1. *$h'_N(t) = \frac{1}{N-1} \cdot \frac{\phi_N(t)}{1-\phi_N(t)}$, so h is strictly increasing.*
2. *$h''_N(t) = \frac{1}{N(N-1)^2\phi_N(t)^{N-2}(1-\phi_N(t))^3}$, so h is strictly convex, $h''(0) = h''(1) = \infty$.*
3. *$h'''_N(t) = \frac{(N+1)\phi_N(t)-(N-2)}{N(N-1)\phi_N(t)^{N-1}(1-\phi_N(t))^2} \cdot h''_N(t)$, so $h'_N(\cdot)$ changes curvature exactly once.*
4. *$f(t) = h_N(t)/t$ satisfies $\lim_{t\to 0} f(t) = 0$ and is strictly increasing on $(0,1)$.*
5. *$h_N(t)$ is increasing in N, for all $t \in (0,1)$ and $N \geq 2$.*

Proof. Parts 1–3 follow immediately from taking derivatives. In Part 4, the first claim follows from the L'Hôpital's rule. For the second claim, notice that by definition of $\phi_N(t)$,

$$\frac{h(t)}{t} = \frac{\phi_N(t)^N}{N\phi_N(t)^{N-1} - (N-1)\phi_N(t)^N} = \frac{1}{\frac{N}{\phi_N(t)} - (N-1)},$$

which is a strictly increasing function of t.

For Part 5, note that ϕ_N is well-defined for any real $N \geq 2$, so we can rely on differentiation. Recall the identity:

$$\left(\alpha(N)^{\beta(N)}\right)'_N = \alpha(N)^{\beta(N)}\beta'(N)\log\alpha(N) + \alpha(N)^{\beta(N)-1}\beta(N)\alpha'(N).$$

Fix any $t \in (0,1)$ and suppress it from the notation for simplicity. Using the above identity:

$$(h_N)' = (\phi_N)^N \log\phi_N + N(\phi_N)^{N-1}(\phi_N)'. \tag{A.1}$$

To compute the derivative $(\phi_N)'$, recall the definition of ϕ_N:

$$t = N(\phi_N)^{N-1} - (N-1)(\phi_N)^N. \tag{A.2}$$

Differentiation both sides using the above identity and rearranging yields:

$$(\phi_N)' = \frac{(\phi_N)^N - (\phi_N)^{N-1} - t\log\phi_N}{N(N-1)(\phi_N)^{N-2}(1-\phi_N)}.$$

Plugging this in (A.1) and using (A.2), with some algebra:

$$\left(\phi^N\right)'_N = \frac{\phi_N}{(N-1)(1-\phi_N)}\left((\phi_N)^N - (\phi_N)^{N-1} - (\phi_N)^{N-1}\log\phi_N\right).$$

The right-hand side is non-negative since $\log(x) \leq x - 1$. ■

Proof of Proposition 1. By Lemma A.1, $u \mapsto \phi_N(u)^N$ is convex. For the lower bound, notice

$$\begin{aligned}
F_{N:N}(v) &= \mathbb{E}[P(V_{N:N} \leq v \,|\, U, N) \,|\, N] \\
&= \mathbb{E}[P(V_j \leq v \,|\, U, N)^N \,|\, N] \\
&= \mathbb{E}[\phi_N(P(V_{N-1:N} \leq v \,|\, U, N))^N \,|\, N] \\
&\geq \phi_N(F_{N-1:N}(v))^N,
\end{aligned}$$

by the law of iterated expectations and Jensen's inequality, for all $v \in \mathcal{V}$. The stated upper bound follows immediately from $V_{N-1:N} \leq V_{N:N}$. ■

Proof of Theorem 1. In the main part of the proof, we show that the stated optimization problems lead to pointwise sharp bounds on $F_{N:N}(v)$. Lemmas C.1 and C.2 show that the bounds are monotone in $F_{N-1:N}(v)$ and thus are plausibly sharp in the functional sense.

For the lower bound, consider the generalized moment problem:

$$\inf_{P \in \mathcal{M}[0,1]} \left\{ \mathbb{E}_P[\phi_N(T)^N] \,\middle|\, \mathbb{E}_P[T] = \mu, c_1 \leq \mathbb{V}ar_P(T) \leq c_2 \right\} \tag{A.3}$$

where $\mathcal{M}[0,1]$ denotes the set of all probability measures supported on $[0,1]$, $\mu \in [0,1]$ and $0 \leq c_1 \leq c_2 \leq \mu(1-\mu)$. Denote $h(t) = \phi_N(t)^N$. Lemma A.1 establishes that $h(t)$ is strictly increasing, strictly convex, and that $h'(t)$ changes curvature exactly once and satisfies $h''(0) = h''(1) = \infty$. Since $h(t)$ is convex, the variance constraint $\mathbb{V}ar_P(T) = c_1$ must be binding, so the problem is:

$$\inf_{P \in \mathcal{M}[0,1]} \left\{ \mathbb{E}_P[h(T)] \,|\, \mathbb{E}_P[T] = \mu, \mathbb{E}_P[T^2] = \mu^2 + c_1 \right\}. \tag{A.4}$$

When $c_1 = 0$, by Jensen's inequality, the infimum is attained by the distribution $P(T = \mu) = 1$ and equals $h(\mu)$. When $c_1 = \mu(1-\mu)$, the only feasible distribution on $[0,1]$ is the Bernoulli distribution with $P(T = 1) = \mu$. In this case, the infimum is equal to μ. It remains to consider $c_1 \in (0, \mu(1-\mu))$.

It is well known that it suffices to consider distributions P with at most three support points (e.g., Theorem 1 in Kemperman, 1968). We first show that, due to the specific shape of the function $h(\cdot)$ and its derivative, it suffices to consider distributions with two support points. Denote $g_1(t) = t$, $g_2(t) = t^2$, and $g(t) = (g_1(t), g_2(t))$ all defined on $\mathcal{T} = [0,1]$. Note that $V = \text{Conv}(g(\mathcal{T})) = \{(z_1, z_2) \in [0,1]^2 : z_1^2 \leq z_2 \leq z_1\}$, and the point $y = (\mu, \mu^2 + c_1) \in \text{Int(V)}$. Denote $D^* = \{d^* = (d_0, d_1, d_2) \in \mathbf{R}^3 : d_0 + d_1 g_1(t) + d_2 g_2(t) \leq h(t) \text{ for all } t \in [0,1]\}$, and, for a given $d^* \in D^*$, let $B(d^*) = \{g(t) \text{ for } t \in \mathcal{T} : d_0 + d_1 g_1(t) + d_2 g_2(t) = h(t)\}$. By Theorem 5 and the following remark in Kemperman (1968), for every $y \in \text{Int}(V)$, there exists a d^* for which $y \in \text{Conv}(B(d^*))$. By Theorem 4 of the same paper, such y can be expressed as $y = \sum_{j=1}^m p_j g(t_j)$ for some $g(t_j) \in B(d^*)$, and the minimal value of the moment problem is given by $\sum_{j=1}^m p_j h(t_j)$. In the problem under consideration, $B(d^*)$ can

contain at most two points. Suppose $k(t) = d_0 + d_1 t + d_2 t^2 \leq h(t)$, for all $t \in \mathcal{T}$, with equality for some $t_1 < \ldots < t_m$. Then: (i) if $t_1 = 0$, there can be at most one other $t_2 \in (0,1)$; (ii) if all $t_j \in (0,1)$, it must be the case that the line $k'(t) = d_1 + 2d_2 t$ intersects the curve $h'(t)$ from above at each t_j.[18] Indeed, case (i) follows from direct computation, and case (ii) from a simple geometric fact that since $h'(t)$ changes curvature only once (concave then convex) and satisfies $h''(0) = h''(1) = \infty$, there are at most two interior intersections of $d_1 + 2d_2 t$ and $h'(t)$. Thus, the set $B(d^*)$ contains at most two points, so to solve (A.4), it suffices to consider distributions P with two support points.

For some $0 \leq a < b \leq 1$ and $p \in [0,1]$, consider the distribution P with $P(X = a) = p$ and $P(X = b) = 1 - p$. The constraints are:

$$\begin{cases} ap + b(1-p) = \mu \\ a^2 p + b^2(1-p) = \mu^2 + c_1 \end{cases} \quad \Longrightarrow \quad \begin{cases} a(p) = \mu - \sqrt{\frac{1-p}{p} c_1} \\ b(p) = \mu + \sqrt{\frac{p}{1-p} c_1}, \end{cases}$$

and p must be subject to $a, b \in [0,1]$. When $p = 0$ or $p = 1$, $a = b = \mu$ which violates the variance constraint for $c_1 > 0$. Thus, the problem is:

$$\min_{p \in (0,1)} \left\{ h\left(\mu - \sqrt{\tfrac{1-p}{p} c_1}\right) p + h\left(\mu + \sqrt{\tfrac{p}{1-p} c_1}\right)(1-p) \right\}$$
$$\text{s.t. } a(p), b(p) \in [0,1]$$

Letting $s = \sqrt{\frac{1-p}{p} c_1}$ so that $p = \frac{c_1}{c_1 + s^2}$ yields an equivalent formulation:

$$\min_{s \in [\frac{c_1}{1-\mu}, \mu]} \left\{ h(\mu - s) \frac{c_1}{c_1 + s^2} + h\left(\mu + \frac{c_1}{s}\right) \frac{s^2}{c_1 + s^2} \right\}.$$

By direct computation, the objective function is seen to be strictly convex on the feasible set.

Next, consider the generalized moment problem:

$$\sup_{P \in \mathcal{M}[0,1]} \left\{ \mathbb{E}_P[h(T)] \;\middle|\; \mathbb{E}_P[T] = \mu, c_1 \leq \mathbb{V}ar_P(T) \leq c_2 \right\},$$

[18] For $t_j \in (0,1)$ it means that $k'(t) > h'(t)$ immediately before t_j and $k'(t) < h'(t)$ immediately after. For $t_j \in \{0,1\}$ only one of the two preceding inequalities is required to hold.

or, equivalently,

$$-\inf_{P\in\mathcal{M}[0,1]}\left\{\mathbb{E}_P[-h(T)]\;\middle|\;\mathbb{E}_P[T]=\mu, c_1\leq\mathbb{V}ar_P(T)\leq c_2\right\}$$

Since $f(t)=-h(t)$ is concave, the variance constraint $\mathbb{V}ar_P(T)=c_2$ must be binding, so the problem is:

$$-\inf_{P\in\mathcal{M}[0,1]}\left\{\mathbb{E}_P[f(T)]\,|\,\mathbb{E}_P[T]=\mu,\mathbb{E}_P[T^2]=\mu^2+c_2\right\}, \tag{A.5}$$

with $\mu\in[0,1]$ and $c_2\in[0,\mu(1-\mu)]$. When $c_2=0$, by the Edmundson-Madansky inequality, the infimum is attained by the distribution $P(T=\mu)=1$ and equals $f(\mu)$. When $c_2=\mu(1-\mu)$, the only feasible distribution on $[0,1]$ is the Bernoulli distribution with $P(T=1)=\mu$. In this case, the infimum is equal to $-\mu$. It remains to consider $c_2\in(0,\mu(1-\mu))$.

Using the idea and notation from the first part of the proof, suppose $k(t)=d_0+d_1t+d_2t^2\leq f(t)$ for all $t\in\mathcal{T}$ with equality for some $t_1<\ldots<t_m$. Then: at each $t_j\in(0,1)$, the line $k'(t)$ must intersect $f'(t)$ from above; if $t_1=0$, it must be $k'(0)\leq f'(0)$; and if $t_m=1$, it must be that $k'(1)\geq f'(1)$. The function $f'(t)$ changes curvature exactly once (convex then concave) and satisfies $f'(0)=0$ and $f'(1)=-\infty$. Thus, there can be at most one interior t_j satisfying the requirement above. By Theorem 4 of Kemperman (1968), it suffices to consider distributions with three support points: 0, 1, and $t\in(0,1)$. Letting $P(T=t)=q$, $P(T=1)=p$, and $P(T=0)=1-p-q$, the constraints in (A.5) become:

$$\begin{cases}p+qt=\mu\\ p+qt^2=\mu^2+c_2\end{cases}\quad\Longrightarrow\quad\begin{cases}q(t)=\frac{\mu(1-\mu)-c_2}{t(1-t)}\\ p(t)=\mu-q(t)\cdot t\end{cases},$$

subject to $q(t),p(t)\geq 0$ and $q(t)+p(t)\leq 1$. Plugging this into (A.5) and rearranging yields:

$$\max_{t\in\left[\mu-\frac{c_2}{1-\mu},\mu+\frac{c_2}{\mu}\right]}\left\{\mu-\frac{\mu(1-\mu)-c_2}{t(1-t)}(t-h(t))\right\}.$$

■

Proof of Proposition 2. Let $h_j(x)=\phi_j(x)^j$, which is increasing in j, by

Lemma A.1. Note that

$$F_1(v) = \sum_{j=\underline{N}}^{\overline{N}} P(N=j)\mathbb{E}[h_j(P(V_{N-1:N} \leq v \,|\, U, N=j)) \,|\, N=j], \tag{A.6}$$

$$F_2(v) = \sum_{j=\underline{N}}^{\overline{N}} P(N=j)\mathbb{E}[P(V_{N-1:N} \leq v \,|\, U, N=j) \,|\, N=j]. \tag{A.7}$$

The lower bound on $F_1(v)$ is given by the solution of the following optimization problem:

$$\min_{p,\{F_j\}_{j=\underline{N}}^{\overline{N}}} \left\{ \sum_{j=\underline{N}}^{\overline{N}} p_j \mathbb{E}_{F_j}[h_j(X_j)] \;\middle|\; \sum_{j=\underline{N}}^{\overline{N}} p_j \mathbb{E}_{F_j}[X_j] = \mu \right\}, \tag{A.8}$$

where F_j denotes the distribution of $X_j = P(V_{N-1:N} \leq v \,|\, U, N=j)$, for each $j \in \{\underline{N}, \ldots, \overline{N}\}$, $p = (p_j)_{j=\underline{N}}^{\overline{N}}$ denotes the distribution of N, and $\mu = F_2(v)$.

Suppose $(p, \{F_j\}_{j=\underline{N}}^{\overline{N}})$ solve (A.8), and $p_j > 0$ for some $j > \underline{N}$. Then, setting $\tilde{p} = (1, 0, \ldots, 0)$, $\tilde{F}_{\underline{N}} = \sum_{k=\underline{N}}^{\overline{N}} p_k F_k$, and $\tilde{F}_j = F_j$, for $j \in \{\underline{N}+1, \ldots, \overline{N}\}$, is feasible and yields a lower value for the objective, since $h_{\underline{N}} \leq h_j$, for any $j \in \{\underline{N}+1, \ldots, \overline{N}\}$. Therefore, the minimum can only be attained at $p = (1, 0, \ldots, 0)$. Thus, (A.8) is equivalent to

$$\min_{F_{\underline{N}}} \left\{ \mathbb{E}_{F_{\underline{N}}}[h_{\underline{N}}(X_{\underline{N}})] \;\middle|\; \mathbb{E}_{F_{\underline{N}}}[X_{\underline{N}}] = \mu \right\}. \tag{A.9}$$

By Lemma A.1, $h_{\underline{N}}$ is convex, so by the Jensen's inequality, the sharp bound is $F_1(v) \geq \phi_{\underline{N}}(F_2(v))^{\underline{N}}$. Arguing similarly for the upper bound, the relevant optimization problem is

$$\max_{F_{\underline{N}}} \left\{ \mathbb{E}_{F_{\underline{N}}}[h_{\underline{N}}(X_{\underline{N}})] \;\middle|\; \mathbb{E}_{F_{\underline{N}}}[X_{\underline{N}}] = \mu; \right\}. \tag{A.10}$$

By the Edmundson-Madansky inequality, the bound $F_1(v) \leq F_2(v)$ is sharp. ■

Proof of Theorem 3. As in the proof of Proposition 2, let $h_j(x) = \phi_j(x)^j$, which is increasing in j, by Lemma A.1, and recall equations (A.6) and (A.7). Note that

$$\mathbb{V}ar(P(V_{N-1:N} \leq v \,|\, U, N)) = \sum_{j=\underline{N}}^{\overline{N}} P(N=j)\mathbb{E}[P(V_{N-1:N} \leq v \,|\, U, N=j)^2 \,|\, N=j] - F_2(v)^2. \tag{A.11}$$

The lower bound on $F_1(v)$ subject to (A.7) and $\mathbb{V}ar(P(V_{N-1:N} \leq v \,|\, U, N)) \geq c$ is obtained by solving the following minimization problem:

$$\min_{p,\{F_j\}_{j=\underline{N}}^{\overline{N}}} \sum_{j=\underline{N}}^{\overline{N}} p_j \mathbb{E}_{F_j}[h_j(X_j)] \quad \text{s.t.} \quad \begin{cases} \sum_{j=\underline{N}}^{\overline{N}} p_j \mathbb{E}_{F_j}[X_j] = \mu; \\ \sum_{j=\underline{N}}^{\overline{N}} p_j \mathbb{E}_{F_j}[X_j^2] \geq \mu^2 + c; \end{cases} \tag{A.12}$$

where F_j denotes the distribution of $X_j = P(V_{N-1:N} \leq v \,|\, U, N = j)$, conditional on $N = j$, for each $j \in \{\underline{N}, \ldots, \overline{N}\}$, $p = (p_j)_{j=\underline{N}}^{\overline{N}}$ denotes the distribution of N, and $\mu = F_2(v)$.

Suppose $(p, \{F_j\}_{j=\underline{N}}^{\overline{N}})$ solve (A.12), and $p_j > 0$ for some $j > \underline{N}$. Then, setting $\tilde{p} = (1, 0, \ldots, 0)$, $\tilde{F}_{\underline{N}} = \sum_{k=\underline{N}}^{\overline{N}} p_k F_k$, and $\tilde{F}_j = F_j$, for $j \in \{\underline{N}+1, \ldots, \overline{N}\}$, is feasible and yields a lower value for the objective, since $h_{\underline{N}} \leq h_j$, for any $j \in \{\underline{N}+1, \ldots, \overline{N}\}$. Therefore, the minimum can only be attained at $p = (1, 0, \ldots, 0)$, which corresponds to setting $P(N = \underline{N}) = 1$. Thus, the problem in (A.12) is equivalent to

$$\min_{F_{\underline{N}}} \mathbb{E}_{F_{\underline{N}}}[h_{\underline{N}}(X_{\underline{N}})] \quad \text{s.t.} \quad \begin{cases} \mathbb{E}_{F_{\underline{N}}}[X_{\underline{N}}] = \mu; \\ \mathbb{E}_{F_{\underline{N}}}[X_{\underline{N}}^2] \geq \mu^2 + c, \end{cases} \tag{A.13}$$

which coincides with the moment problem considered in the proof of Theorem 1, so the rest of the argument follows from that proof. Remark that the constant c effectively restricts the variance of $P(V_{N-1:N} \leq v \,|\, U, N = \underline{N})$. The proof for the upper bound, subject to $\mathbb{V}ar(P(V_{N-1:N} \leq v \,|\, U, N)) \leq c$, is symmetric. ■

Before proving the Min-Max-Regret theorems, we state an auxiliary result. Its' proof additionally shows that $V_{N-1:N}$ in the statement of Assumption 3.2 can be replaced with $V_{N-1:N-1}$.

Lemma A.2. *Suppose $(V_1, \ldots, V_N)$ are exchangeable, and Assumption 3.2 holds. Suppose the marginal CDF $F(v)$ of V_j is strictly increasing. Then, $F_{N:N} = g_N(F_{N-1:N})$, for some increasing convex function $g_N : [0,1] \to [0,1]$.*

Proof. Let $f_V = F'$ denote the density function of V_j. Let $\mathcal{C}_J : [0,1] \to [0,1]$ be defined as $\mathcal{C}_J(u) = \tilde{\mathcal{C}}_J(u, \ldots, u)$ where $\tilde{\mathcal{C}}_J$ is the copula function of $(V_1, \ldots, V_J)$, for

$J \in \{N-1, N\}$. By exchangeability,

$$\begin{aligned} F_{N-1:N}(v) &= N \cdot \mathcal{C}_{N-1}(F(v)) - (N-1) \cdot \mathcal{C}_N(F(v)) \\ F_{N:N}(v) &= \mathcal{C}_N(F(v)). \end{aligned}$$

Denote $f(u) = N\mathcal{C}_{N-1}(u) - (N-1)\mathcal{C}_N(u)$ and notice this function is strictly increasing. Then,

$$F_{N:N}(v) = \mathcal{C}_N(f^{-1}(F_{N-1:N}(v))) \equiv g(F_{N-1:N}(v)).$$

By direct calculation:

$$g''(u) = \frac{\mathcal{C}_N''(s_u)f'(s_u) - \mathcal{C}_N'(s_u)f''(s_u)}{f'(s_u)^3},$$

where $s_u = f^{-1}(u)$. Plugging in the expressions for $f'(s_u)$ and $f''(s_u)$ and simplifying yields that $g''(u) \geq 0$, for all $u \in [0,1]$, holds if and only if:

$$\frac{\mathcal{C}_N''(s)}{\mathcal{C}_N'(s)} \geq \frac{\mathcal{C}_{N-1}''(s)}{\mathcal{C}_{N-1}'(s)},$$

for all $s \in [0,1]$. Denoting $c_J(s) = \mathcal{C}_J'(s)$ for $J \in \{N-1, N\}$, the above is equivalent to:

$$\frac{\partial}{\partial s}\left(\frac{c_N(s)}{c_{N-1}(s)}\right) \geq 0.$$

Since $f_{V_{N:N}}(v) = c_N(F(v))f_V(v)$ and $f_{V_{N-1:N}}(v) = (Nc_{N-1}(F(v)) - (N-1)f_{V_{N:N}}(v))f_V(v)$, it follows that:

$$\frac{f_{V_{N-1:N}}(v)}{f_{V_{N:N}}(v)} = N\frac{c_{N-1}(F(v))}{c_N(F(v))} - (N-1),$$

so $\frac{f_{V_{N:N}}(v)}{f_{V_{N-1:N}}(v)}$ is increasing if and only if $\frac{c_N(F(v))}{c_{N-1}(F(v))}$ is, which concludes the proof. ■

Proof of Theorem 5. Let F denote $F_{N-1:N}$ and F^* denote $F_{N:N}$, for simplicity. Let $\pi(r; F^*)$ denote the expected profit function in (2), and $\mathcal{F}$ the set of all admissible F^*. Fix some reserve $r \geq v_0$ and consider an alternative $\tilde{r} \geq v_0$. Denote $D(\tilde{r}, r; F^*) = \pi(\tilde{r}; F^*) - \pi(r; F^*)$, which is

$$D(\tilde{r}, r; F^*) = \int_{\tilde{r}}^{r} (1 - F(v))dv + \tilde{r} - r + \{F^*(r)(r - v_0) - F^*(\tilde{r})(\tilde{r} - v_0)\}.$$

The maximum regret is given by $R(r) = \max\{R_1(r), R_2(r)\}$, where

$$R_1(r) = \sup_{F^*} \sup_{\tilde{r} \leq r} D(\tilde{r}, r; F^*),$$

$$R_2(r) = \sup_{F^*} \sup_{\tilde{r} > r} D(\tilde{r}, r; F^*).$$

Considering $D(\tilde{r}, r; F^*)$, the supremum $R_1(r)$ is attained by (i) maximizing $F^*(r)$; and (ii) minimizing $F^*(\tilde{r})$ for all $\tilde{r} < r$. Both conditions are satisfied by the CDF $F^*_{1,r}$ in the statement of the theorem. Similarly, the supremum $R_2(r)$ is attained by: (i) maximizing $F^*(r)$; and (ii) minimizing $F^*(\tilde{r})$ for all $\tilde{r} > r$. Condition (i) is achieved by setting $F^*(r) = \overline{F}(r)$. Since F^* must be nondecreasing, condition (ii) is achieved by setting $F^*(v) = \overline{F}(r)$ for all $v \in [r, \underline{F}^{-1}(\overline{F}(r))]$ and to $F^*(v) = \underline{F}(v)$ afterwards. Both of these conditions are satisfied by the CDF $F^*_{2,r}$ in the statement of the theorem. Hence, the result follows. ■

Proof of Theorem 6. The argument is essentially the same as in Theorem 5. The supremum $R_1(r)$ is attained by (i) maximizing $F_1(r)$; and (ii) minimizing $F_1(\tilde{r})$ for all $\tilde{r} < r$. Since it must be that $F_1(v) = g_1(F_2(v))$ for a convex function g_1, conditions (i) and (ii) can only be attained by defining $g_1 = g_{1,r}$ as in the statement of the Theorem. The middle part of $g_{1,r}$ is restricted to be linear precisely to preserve convexity while minimizing $F_1(\tilde{r})$ for all $\tilde{r} < r$. A similar argument applies for the supremum $R_2(r)$ and $g_{2,r}$. ■

Online Appendix

B Calibration

Below, we discuss a procedure to calibrate ρ, which captures the strength of the dependence between $V_{N-1:N}$ and $\tilde{V}_{N-1:N}$ through unobserved heterogeneity U in Assumption 2.4. We start with a formal result.

Lemma B.1. *Let Assumption 2.2 hold and suppose the reserve prices are not binding, so that the transaction price equals the second largest valuation, $P = V_{N-1:N}$. Let Assumption 2.3 hold with valuations being additive, $V_j = U + \varepsilon_j$, for $j \in \{1, \ldots, N\}$. Partition the observed covariates as $X_L = (X_S, X_R)$, where X_S is the short list and X_R contains the remaining covariates. Define the following variables*

$$\rho = \frac{\mathbb{V}ar(U - \mathbb{E}[U \mid X_S])}{\mathbb{V}ar(P - \mathbb{E}[P \mid X_S])},$$

$$\mathrm{MSE}_{\mathrm{long}} = \mathbb{E}\big[(P - \mathbb{E}[P \mid X_S, X_R])^2\big],$$

$$\mathrm{MSE}_{\mathrm{short}} = \mathbb{E}\big[(P - \mathbb{E}[P \mid X_S])^2\big].$$

Here, ρ captures the strength of dependence between $V_{N-1:N}$ and $\tilde{V}_{N-1:N}$ conditional on the set of covariates X_S. Then:

(i) (No Missing Covariates.) *If $U = f(X_S, X_R)$, where f is a measurable function, and $\varepsilon_{N-1:N} \perp (X_S, X_R)$, then*

$$1 - \frac{\mathrm{MSE}_{\mathrm{long}}}{\mathrm{MSE}_{\mathrm{short}}} = \rho.$$

(ii) (Missing covariates.) *If $U = f(X_S, X_R, U_0)$, for some unobserved U_0 and measurable function f, and $\varepsilon_{N-1:N} \perp (X_S, X_R, U_0)$, then*

$$1 - \frac{\mathrm{MSE}_{\mathrm{long}}}{\mathrm{MSE}_{\mathrm{short}}} \leq \rho.$$

Proof. By the imposed assumptions, we can write $P = U + \varepsilon$, where we denote $\varepsilon = \varepsilon_{N-1:N}$, to simplify the notation.

We prove (ii) first. Denote $\mathbb{E}[\varepsilon \,|\, X_S, X_R, U_0] = \mu_\varepsilon = const.$ By the law of iterated expectations, $\mathbb{E}[\varepsilon \,|\, X_S, X_R] = \mu_\varepsilon$. As a result,

$$\begin{aligned} \mathbb{E}[P \,|\, X_S, X_R] &= \mathbb{E}[U \,|\, X_S, X_R] + \mu_\varepsilon \\ \mathbb{E}[P \,|\, X_S] &= \mathbb{E}[U \,|\, X_S] + \mu_\varepsilon. \end{aligned}$$

Let $\nu = U - \mathbb{E}[U \,|\, X_S, X_R]$ and $\eta = \mathbb{E}[U \,|\, X_S, X_R] - \mathbb{E}[U \,|\, X_S]$, so that $U - \mathbb{E}[U \,|\, X_S] = \nu + \eta$. Then,

$$\begin{aligned} P - \mathbb{E}[P \,|\, X_S, X_R] &= \nu + (\varepsilon - \mu_\varepsilon), \\ P - \mathbb{E}[P \,|\, X_S] &= \nu + \eta + (\varepsilon - \mu_\varepsilon). \end{aligned}$$

The terms ν, η, $\varepsilon - \mu_\varepsilon$ are mean zero and mutually uncorrelated. As a result,

$$\begin{aligned} \text{MSE}_{\text{long}} &= \mathbb{V}ar(\nu) + \mathbb{V}ar(\varepsilon), \\ \text{MSE}_{\text{short}} &= \mathbb{V}ar(\nu) + \mathbb{V}ar(\eta) + \mathbb{V}ar(\varepsilon), \\ \rho &= \frac{\mathbb{V}ar(\nu) + \mathbb{V}ar(\eta)}{\mathbb{V}ar(\nu) + \mathbb{V}ar(\eta) + \mathbb{V}ar(\varepsilon)}. \end{aligned}$$

Hence,

$$1 - \frac{\text{MSE}_{\text{long}}}{\text{MSE}_{\text{short}}} = \frac{\mathbb{V}ar(\eta)}{\mathbb{V}ar(\nu) + \mathbb{V}ar(\eta) + \mathbb{V}ar(\varepsilon)} \leq \frac{\mathbb{V}ar(\nu) + \mathbb{V}ar(\eta)}{\mathbb{V}ar(\nu) + \mathbb{V}ar(\eta) + \mathbb{V}ar(\varepsilon)} = \rho$$

To prove (i), notice $U = f(X_S, X_R)$ gives $\mathbb{E}[U \,|\, X_S, X_R] = U$, so $\nu = 0$ and the inequality in the above display holds with equality. ■

B.1 Implementation

Lemma B.1 states that $1 - \text{MSE}_{\text{long}}/\text{MSE}_{\text{short}}$ is an exact specification of the dependence parameter if the set of long covariates fully span the common component, and otherwise, it is a lower bound.

We compute MSE_{long} and $\text{MSE}_{\text{short}}$ as out of sample mean-squared errors of two prediction models. Table 5 lists the covariates that enter the short and long models. We evaluate ρ on six subsamples of sold lots, all of which are impressionist/20th–21st-century art sold in New York. The subsamples are defined by the presale high-estimate interval, [100K, 1M] or [1M, 10M], and the number of bidders interval, $\underline{N} \in [2, 5]$,

Long covariates, X_L	**Description**
auction house	Christie's or Sotheby's
location	New York, Hong Kong, Las Vegas, London, Shanghai
category of art	Chinese, Impressionist/20-21 Century, Old Masters
low estimate / high estimate	the auction house's measure of uncertainty
high estimate	auction house's high estimate
lot number	the lot's position within the auction
close date	auction date
is signed	whether the work is signed by the artist
dimensions	dimensions of the art / sculpture
count past owners	number of previous owners (from provenance)
count past auctions	number of previous auctions (from provenance)
count literature	number of literature references
count times exhibited	number of exhibitions (from exhibited)
minimum-price guarantee	whether the work has a minimum-price guarantee
third-party guarantee	whether the work has a third-party guarantee
lower bound on N	lower bound on the number of bidders
Short covariates, X_S	
high-estimate indicator	100K–1M or 1M–10M
lower bound on bidders indicator	2–5, 6–20, or 2–20
category of art	Chinese, Impressionist/20-21 Century, Old Masters
location	New York, Hong Kong, Las Vegas, London, Shanghai

Table 5: Predictors in the long and short models.

$\underline{N} \in [6, 20]$, or $\underline{N} \in [2, 20]$.

We use random forests, which often predict well off the shelf. For each of the six subsamples, we fit two random forests on all lots outside the subsample, one with long covariates X_L, and one with short covariates X_S. We set the same hyperparameters for both models: The *number of trees* in the forest is 300; The *maximum depth* to which each tree is grown is 16; The *minimum leaf size*, i.e., the fewest training lots allowed in a leaf is 10; The number of *features per split*, the fraction of covariates drawn at each split: 1/2.

In our statistical analysis, the bounds $\underline{\rho}, \overline{\rho}$ are assumed to be known and fixed. To comply with this assumption, we draw a random holdout of 30 lots from each subsample and compute out-of-sample MSE_{long} and $\text{MSE}_{\text{short}}$ on that holdout to obtain the estimates of $\underline{\rho}, \overline{\rho}$. The remaining part of each subsample is used for estimating the MM and MMR reserves and the corresponding bounds on the profit function.

C Auxiliary Results in The Main Text

C.1 Properties of the Bounds in Theorem 1

Lemma C.1. *Consider the function:*

$$f(\mu) = \min_{s\in[\frac{c_1}{1-\mu},\mu]} \left\{ \phi_N(\mu - s)^N \frac{c_1}{c_1 + s^2} + \phi_N \left(\mu + \frac{c_1}{s}\right)^N \frac{s^2}{c_1 + s^2} \right\}$$

with $c_1 = C_\rho(\mu,\mu) - \mu^2$ with C_ρ satisfying the requirements in Assumption 2.4. Then, $\mu \mapsto f(\mu)$ is nondecreasing.

Proof. We suppress the dependence of c_1 on μ to simplify the notation. Denoting the objective function in the optimization problem by $r(s;\mu,c_1)$,

$$\frac{\partial r(s;\mu,c_1)}{\partial s} = \frac{2c_1 s}{(c_1+s^2)^2}\left\{h\left(\mu+\frac{c_1}{s}\right) - h(\mu-s)\right\} - \frac{c_1}{c_1+s^2}\left\{h'(\mu-s) + h'\left(\mu+\frac{c_1}{s}\right)\right\}.$$

First, note that as $s \to \frac{c_1}{1-\mu}$, we have $\mu + c_1/s \to 1$ and thus $h'(\mu + c_1/s) \to +\infty$, by Lemma A.1. Since all other terms in the above expression stay finite, $r'(s;\mu,c_1) \to -\infty$. Thus, the constraint $s \geq c_1/(1-\mu)$ never binds, and it suffices to consider two cases: (i) $s = \mu$ and (ii) $s \in (c_1/(1-\mu), \mu)$. In case (i), the lower bound is given by:

$$f(\mu) = \frac{\mu^2}{c_1+\mu^2} h\left(\mu + \frac{c_1}{\mu}\right) = \mu \frac{h(\frac{C_\rho(\mu,\mu)}{\mu})}{\frac{C_\rho(\mu,\mu)}{\mu}}.$$

The function $u \mapsto h(u)/u$ is increasing, by Lemma A.1, and $C_\rho(\mu,\mu)/\mu$ is increasing by Assumption 2.4, so $f(\mu)$ is increasing. For case (ii), by the Envelope Theorem:

$$f'(\mu) = \left.\frac{\partial r(s,\mu,c_1)}{\partial \mu}\right|_{s=s^*(\mu)} + \left.\frac{\partial r(s;\mu,c_1)}{\partial c_1}\right|_{s=s^*(\mu)} \cdot \frac{\partial c_1}{\partial \mu}, \tag{C.1}$$

where $s^*(\mu)$ solves the first-order condition $\frac{\partial r(s;\mu,c_1)}{\partial s} = 0$,

$$\frac{\partial r(s,\mu,c_1)}{\partial \mu} = \frac{c_1}{c_1+s^2} h'(\mu-s) + \frac{s^2}{c_1+s^2} h'\left(\mu+\frac{c_1}{s}\right),$$

and

$$\frac{\partial r(s;\mu,c_1)}{\partial c_1} = \frac{s^2}{(c_1+s^2)^2}\left\{h(\mu-s) - h\left(\mu+\frac{c_1}{s}\right)\right\} + \frac{s}{c_1+s^2}h'\left(\mu+\frac{c_1}{s}\right). \tag{C.2}$$

Thus, from Equation C.1, the condition $f'(\mu) \geq 0$ is equivalent to:

$$\frac{\partial c_1(\mu)}{\partial \mu} \geq -\left.\frac{\frac{\partial r(s,\mu,c_1)}{\partial \mu}}{\frac{\partial r(s;\mu,c_1)}{\partial c_1}}\right|_{s=s^*(\mu)} \tag{C.3}$$

From the first-order condition, at $s = s^*(\mu)$,

$$h\left(\mu+\frac{c_1}{s}\right) - h(\mu-s) = \frac{c_1+s^2}{2s}\left\{h'(\mu-s) + h'(\mu+\frac{c_1}{s})\right\}.$$

Plugging the above in (C.2) and the result in (C.3), and applying the Mean Value Theorem yields:

$$\begin{aligned}
-\left.\frac{\frac{\partial r(s,\mu,c_1)}{\partial \mu}}{\frac{\partial r(s;\mu,c_1)}{\partial c_1}}\right|_{s=s^*(\mu)} &= -2\left.\frac{h'(\mu-s) + \frac{s^2}{c_1+s^2}\{h'(\mu+\frac{c_1}{s}) - h'(\mu-s)\}}{\frac{s}{c_1+s^2}\{h'(\mu+\frac{c_1}{s}) - h'(\mu-s)\}}\right|_{s=s^*(\mu)} \\
&= -2\left.\frac{h'(\mu-s) + sh''(\tilde{\mu})}{h''(\tilde{\mu})}\right|_{s=s^*(\mu)} \\
&= -2(s^*(\mu) + \tfrac{h'(\mu-s^*(\mu))}{h''(\tilde{\mu})}),
\end{aligned}$$

for some $\tilde{\mu} \in [\mu - s^*(\mu), \mu + \frac{c_1}{s^*(\mu)}]$. Since both $h'(\cdot)$ and $h''(\cdot)$ are positive, and $s^*(\mu) \geq c_1/(1-\mu)$ a sufficient condition for the lower bound to increase in μ is:

$$\frac{\partial c_1(\mu)}{\partial \mu} \geq -2\frac{c_1(\mu)}{1-\mu}.$$

Plugging in the expression for $c_1(\mu)$ and simplifying yields $(C_\rho(\mu,\mu))' \geq 2 \cdot \frac{\mu - C_\rho(\mu,\mu)}{1-\mu}$.
■

Lemma C.2. *Consider the function:*

$$f(\mu) = \max_{s\in\left[\mu-\frac{c_2}{1-\mu},\mu+\frac{c_2}{\mu}\right]} \left\{\mu - \frac{\mu(1-\mu)-c_2}{s(1-s)}(s-h(s))\right\}.$$

where $h(s) = \phi_N(s)^N$ *and* $c_2 = C_\rho(\mu,\mu) - \mu^2$ *with* C_ρ *satisfying the requirements in Assumption 2.4. Then,* $\mu \mapsto f(\mu)$ *is nondecreasing.*

Proof. Let $s^*(\mu)$ denote the argmax. There are three possible cases.

Case (i): Suppose $s^*(\mu) = \frac{\mu - C_\rho(\mu,\mu)}{1-\mu}$. Write $s = s^*(\mu)$ and define

$$A(s) = \frac{1 - h(s)}{1 - s}.$$

The objective simplifies to $f(\mu) = 1 - (1-\mu)A(s)$. By convexity of h and $h(1) = 1$,

$$A'(s) = \frac{A(s) - h'(s)}{1 - s} \geq 0.$$

Moreover, Assumption 2.4(iii) implies

$$(1-\mu)s'(\mu) = 1 + s - \frac{\partial C_\rho(\mu,\mu)}{\partial \mu} \leq 1 - s,$$

Therefore,

$$\begin{aligned} f'(\mu) &= A(s) - (1-\mu)s'(\mu)A'(s) \\ &\geq A(s) - (1-s)A'(s) \\ &= h'(s) \geq 0. \end{aligned}$$

Hence, f is non-decreasing.

Case (ii): $s^*(\mu)$ is interior. Denoting the objective function in the optimization problem by $r(s;\mu,c_2)$, by the Envelope Theorem:

$$\begin{aligned} f'(\mu) &= \left.\frac{\partial r(s,\mu,c_2)}{\partial \mu}\right|_{s=s^*(\mu)} + \left.\frac{\partial r(s;\mu,c_2)}{\partial c_2}\right|_{s=s^*(\mu)} \cdot \frac{\partial (C_\rho(\mu,\mu) - \mu^2)}{\partial \mu} \\ &= 1 - (1-2\mu)\frac{s^* - h(s^*)}{s^* - s^{*2}} + \frac{s^* - h(s^*)}{s^* - s^{*2}} \cdot \left(\frac{\partial C_\rho(\mu,\mu)}{\partial \mu} - 2\mu\right) \\ &= 1 + t(s^*)\left(\frac{\partial C_\rho(\mu,\mu)}{\partial \mu} - 1\right), \end{aligned}$$

where $t(s^*) \equiv \frac{s^* - h(s^*)}{s^* - s^{*2}}$. Notice the solution s^* does not depend on μ and corresponds to the minimum of the function $t(s)$. Since $\frac{\partial C_\rho(\mu,\mu)}{\partial \mu} \geq 0$, to conclude that $f'(\mu) \geq 0$ it suffices to show that $t(s^*) \in [0,1]$. Since $h(s) \leq s$, we have that $t(s) \geq 0$ for all

s. To establish $t(s) = 1 + \frac{s^2 - h(s)}{s - s^2} \leq 1$, it suffices to show $s^2 - h(s) \leq 0$ for some s. By the Taylor expansion, $h(s) = \frac{h''(\tilde{s})}{2} s^2$, for some $\tilde{s} \in (0, s)$, and, by Lemma A.1, $h''(\tilde{s}) \to +\infty$ as $\tilde{s} \to 0$. Thus, for s close to zero, $s^2 - h(s) < 0$, and the result follows.

Case (iii): $s^*(\mu) = \mu + \frac{c_2}{\mu} = \frac{C_\rho(\mu,\mu)}{\mu}$. Plugging this in the objective function and simplifying yields:

$$f(\mu) = \mu \frac{h\left(\frac{C(\mu,\mu)}{\mu}\right)}{\left(\frac{C(\mu,\mu)}{\mu}\right)}.$$

Since $\mu \mapsto \frac{C(\mu,\mu)}{\mu}$ is increasing, by Assumption 2.4, and $t \mapsto \frac{h(t)}{t}$ is increasing, by Lemma A.1, it follows that $f(\mu)$ is increasing.

■

C.2 Properties of the Copula Functions

Below, we present two auxiliary lemmas about the properties of the copula functions that are used to bound the variances $D_N(v)$.

Lemma C.3. *Let U_1, U_2 denote continuous random variables, supported on $[0,1]$, marginally distributed as $U_j \sim U[0,1]$, for $j \in \{1,2\}$, with a joint CDF $C : [0,1]^2 \to [0,1]$. Suppose C is continuously differentiable and $C(u_1, u_2) = C(u_2, u_1)$, for all $u_1, u_2 \in [0,1]$. Define $c(u) = C(u,u)$. Then:*

1. *$c'(u) \geq \frac{c(u)}{u}$, $\forall\, u \in [0,1]$, if and only if $u \mapsto P(U_1 \leq u \,|\, U_2 \leq u)$ is nondecreasing.*
2. *$c'(u) \geq 2\frac{u - c(u)}{1-u}$, $\forall\, u \in [0,1]$, if and only if $P(U_1 \leq u \,|\, U_2 = u) \geq P(U_1 \leq u \,|\, U_2 > u)$.*

Proof. For the first statement, note that $P(U_1 \leq u \,|\, U_2 \leq u) = \frac{c(u)}{u}$, which is nondecreasing if and only if

$$\left(\frac{c(u)}{u}\right)' = \frac{c'(u)u - c(u)}{u^2} \geq 0 \iff c'(u) \geq \frac{c(u)}{u}.$$

For the second statement, let $C_{12}(u_1, u_2)$ denote the joint density of (U_1, U_2), and $C_2(u_1, u_2) = \partial C(u_1, u_2)/\partial u_2$.

$$P(U_1 \leq u \,|\, U_2 = u) = \int_0^u C_{12}(u_1, u) du_1 = C_2(u, u)$$

Using the symmetry of C,

$$c'(u) = 2C_2(u,u) = 2P(U_1 \leq u \,|\, U_2 = u).$$

On the other hand,

$$P(U_1 \leq u \,|\, U_2 > u) = \frac{P(U_1 \leq u) - P(U_1 \leq u, U_2 \leq u)}{1 - P(U_2 \leq u)} = \frac{u - c(u)}{1-u}.$$

Thus, the stated result follows.

■

Lemma C.4. *The functions (i) $C_\rho(u) = \rho u + (1-\rho)u^2$; (ii) $C_\rho(u) = u^{2-\rho}$; and (iii) $C_\rho(u) = P(\Phi(Z_1) \leq u, \Phi(Z_2) \leq u)$ where (Z_1, Z_2) are jointly normal with zero means, unit variances, and covariance ρ, satisfy all requirements of Assumption 2.4.*

Proof. Recall from the proof of Lemma C.3 that $(C_\rho(u))' \geq \frac{C(u)}{u}$ is equivalent to $\frac{C_\rho(u)}{u}$ being increasing in u.

(i) $C_0(u) = u^2$ and $C_1(u) = u$ is immediate. Further, $\frac{C_\rho(u)}{u} = \rho + (1-\rho)u$, which is increasing in u. Finally,

$$2\frac{u - C_\rho(u)}{1-u} = 2u(1-\rho) \leq \rho + 2u(1-\rho) = (C_\rho(u))',$$

(ii) $C_0(u) = u^2$ and $C_1(u) = u$ is immediate. Further, $\frac{C_\rho(u)}{u} = u^{1-\rho}$, which is increasing in u. Finally, consider a function

$$g(u) = (C_\rho(u))' - 2\frac{u - C_\rho(u)}{1-u} = \frac{u^{1-\rho}}{1-u}\left\{2 - \rho + \rho u - 2u^\rho\right\}.$$

It is straightforward to verify that the function in the curly brackets is non-negative.

(iii) As is well known:

$$C_\rho(u) = 2\int_0^u \Phi\left(\sqrt{\frac{1-\rho}{1+\rho}}\Phi^{-1}(t)\right) dt.$$

Hence, $C_0(u) = u^2$ and $C_1(u) = u$ is immediate. Further, since the integrand in the above display is non-negative and increasing in t,

$$C_\rho(u) \leq 2\Phi\left(\sqrt{\frac{1-\rho}{1+\rho}}\Phi^{-1}(u)\right) u$$

By the Leibniz rule,

$$(C_\rho(u))' = 2\Phi\left(\sqrt{\frac{1-\rho}{1+\rho}}\Phi^{-1}(u)\right),$$

so the inequality in the preceding display states $(C_\rho(u))' \geq \frac{C_\rho(u)}{u}$.

It remains to show $(C_\rho(u))' \geq 2\frac{u-C_\rho(u)}{1-u}$. Let $z = \Phi^{-1}(u)$ and ϕ denote the standard normal density. Note that

$$\begin{aligned}
u - C_\rho(u) &= P(Z_1 \leq z) - P(Z_1 \leq z, Z_2 \leq z) \\
&= P(Z_1 \leq z, Z_2 > z) \\
&= \int_z^\infty \Phi\left(\tfrac{z-\rho t}{\sqrt{1-\rho^2}}\right)\phi(t)dt \\
&\leq \Phi\left(\tfrac{z-\rho z}{\sqrt{1-\rho^2}}\right)\int_z^\infty \phi(t)dt \\
&= \Phi\left(\sqrt{\tfrac{1-\rho}{1+\rho}}\Phi^{-1}(u)\right)(1-u) \\
&= (C_\rho(u))'\tfrac{1-u}{2},
\end{aligned}$$

where the inequality follows from the fact that $t \geq z$ under the integral and $\rho \geq 0$. Re-arranging yields the stated result.

■

C.3 Relaxed MMR and Max-Min Reserves

In this section, we show that under mild conditions the relaxed MMR and max-min solutions coincide. Suppose the bounds on $F_{N:N}(v)$ take the form:

$$\underline{F}(v) \leq F_{N:N}(v) \leq \overline{F}(v),$$

for all $v \in \mathcal{V}$, where $\overline{F}(v) \leq F_{N-1:N}(v)$. Let

$$\underline{\pi}(s) = \pi_N(s; F_{N-1:N}, \overline{F}); \qquad \overline{\pi}(s) = \pi_N(s; F_{N-1:N}, \underline{F})$$

denote the corresponding profit bounds, which we will assume are single-peaked. Let

$$\underline{r}^* = \operatorname*{argmax}_{s \geq v_0} \underline{\pi}(s) \qquad \overline{r}^* = \operatorname*{argmax}_{s \geq v_0} \overline{\pi}(s)$$

denote the maximizers of the lower and upper profit bounds correspondingly. Notice that the max-min reserve is $\underline{r}^*$. Define $\underline{g} = \max\{r \in \mathcal{V} : \overline{F}(r) \leq \underline{F}(\overline{r}^*)\}$ and $W(r) = \overline{\pi}(r) - \underline{\pi}(r)$.

Proposition 3. *Let Assumptions 2.1, 2.2, and 3.1 hold, and the functions $\underline{\pi}, \overline{\pi}$ be single-peaked. If $\underline{r}^* \leq \underline{g}$ and $W(r) \geq \overline{\pi}(\overline{r}^*) - \underline{\pi}(\underline{r}^*)$ for all $r \in \left(\underline{g}, \overline{r}^*\right)$, then the relaxed MMR reserve r^* is equal to the MM reserve $\underline{r}^*$.*

Proof. From Theorem 5, we have

$$R_1(r) = \sup_{s \in [v_0, r]} \overline{\pi}(s) - \underline{\pi}(r)$$

and

$$R_2(r) = \sup_{s \geq \overline{v}(r)} \overline{\pi}(s) - \underline{\pi}(r),$$

since $\pi_N(s; F_{N-1:N}, F_{2,r})$ is increasing on the interval $[r, \overline{v}(r)]$. Thus, we can write the maximum regret as

$$R(r) = \sup_{s \in [v_0, r] \cup [\overline{v}(r), \infty)} \overline{\pi}(s) - \underline{\pi}(r).$$

We will consider two cases:

1. First, consider $r \notin \left(\underline{g}, \overline{r}^*\right)$. Here, $\overline{r}^* \in [v_0, r] \cup [\overline{v}(r), \infty)$, by construction, and thus $R(r) = \overline{\pi}(\overline{r}^*) - \underline{\pi}(r)$. Since $\underline{r}^* \leq \underline{g}$ and thus $\underline{r}^* \notin (\underline{g}, \overline{r}^*)$ by assumption, $R(r)$ is minimized at $\underline{r}^*$.

2. Next, consider $r \in \left(\underline{g}, \overline{r}^*\right)$. Here, since $r \in [v_0, r] \cup [\overline{v}(r), +\infty)$, we have $R(r) \geq \overline{\pi}(r) - \underline{\pi}(r) = W(r) \geq \overline{\pi}(\overline{r}^*) - \underline{\pi}(\underline{r}^*) = R(\underline{r}^*)$.

Thus, $R(r)$ is minimized at $\underline{r}^*$, so that the relaxed MMR and max-min solutions coincide. ■

D Estimation

For brevity, the discussion below treats N as known and fixed, and $X = \varnothing$. Extensions to unknown or varying N and discrete X are straightforward. Continuous covariates require a different type of asymptotic approximation, which is beyond the scope of this paper. Let $S_1, \ldots, S_n \sim P$ be a random sample of selling prices, and denote $F_P(v) = P(S \leq v)$. We focus on the empirical CDF estimator, $\hat{F}(v) = n^{-1} \sum_{i=1}^{n} \mathbf{1}(S_i \leq v)$, but other choices, e.g., the kernel-smoothed estimators, are easily accommodated.[19]

For a set $\mathcal{T}$, let $\ell^\infty(\mathcal{T}) = \{h : \mathcal{T} \to \mathbf{R} : \sup_{t \in \mathcal{T}} |h(t)| < \infty\}$. For any $h \in \ell^\infty(\mathcal{T})$, let $||h||_{\mathcal{T}} = \sup_{t \in \mathcal{T}} |h(t)|$. We will be interested primarily in $\ell^\infty(\mathcal{F})$, for a class of functions $\mathcal{F}$ defined below, and $\ell^\infty(\mathcal{R})$, where $\mathcal{R} \subseteq \mathbf{R}$ denotes the set of plausible reserve prices. Let $C(\mathcal{R}) \subseteq \ell^\infty(\mathcal{R})$ be the class of all continuous functions $f : \mathcal{R} \to \mathbf{R}$, also endowed with the sup-norm. Let $\rightsquigarrow$ denote weak convergence. Recall that a function $f : \mathcal{R} \to \mathbf{R}$ is *strictly single-peaked* if it is strictly increasing for all $r \leq r^*$ and strictly decreasing for all $r \geq r^*$, for some $r^* \in \mathcal{R}$.

We establish the validity of asymptotic approximations and the bootstrap procedure under the following assumption.

Assumption D.1.

(1) $\mathcal{R} = [v_0, \overline{r}]$, *for some* $\overline{r} < \infty$.

(2) The transaction price S *satisfies* $\mathbb{E}_P[S^2] < \infty$.

(3) All functions in the class $\mathcal{G} = \{\underline{\psi}, \overline{\psi}\} \cup \{g_{1,r} \circ \underline{\psi}, g_{2,r} \circ \underline{\psi}, g_{1,r} \circ \overline{\psi}, g_{2,r} \circ \overline{\psi}\}_{r \in \mathcal{R}}$, *where* $g_{1,r}, g_{2,r}$ *are defined in Theorem 6, are continuously differentiable.*

(4) The function $s \mapsto \pi(s; P, g)$ *is continuous and strictly single-peaked, for* $g \in \{g_{1,r}, g_{2,r}\}$, *for all* $r \in \mathcal{R}$, *for* $g_{1,r}, g_{2,r}$ *as defined in Theorem 6.*

(5) The MMR reserve $r^*(P)$ *uniquely solves* $\Delta(r^*(P); P) = 0$. *The function* $r \mapsto \Delta(r; P)$ *is continuously differentiable, and* $\Delta'(r^*; P) \neq 0$.

[19] The formal argument is virtually the same as the one provided below with the class $\mathcal{F}_2$ defined instead as $\mathcal{F}_2 = \{x \mapsto K(\frac{y-x}{h}) : y \in \mathbf{R}, h > 0\}$, where $K(\cdot)$ is a fixed continuous CDF. This class is VC-subgraph and pointwise measurable, and thus Donsker. As a result, kernel-smoothed estimator converges weakly when recentered around the smoothed target. In turn, weak convergence of the kernel-smoothed CDF estimator to the true CDF requires undersmoothing, $\sqrt{n}h_n^2 \to 0$.

Conditions (1)–(3) are mild. Conditions (4) and (5) may be restrictive but they are essentially necessary for the validity of the plug-in bootstrap procedure. Under Assumption D.1, we show that the map $P \mapsto r^*(P)$ is Hadamard differentiable, which is a necessary and sufficient condition for the validity of plug-in bootstrap (Fang and Santos, 2019). Specifically, when either D.1(3) or D.1(4) are violated, Hadamard differentiability generally fails and plug-in bootstrap is invalid.

Define the class of functions $\mathcal{F} = \mathcal{F}_1 \cup \mathcal{F}_2$, where $\mathcal{F}_1 = \{f : \mathbf{R} \to \mathbf{R} : f(v) = \max(v, r) : r \in \mathcal{R}\}$ and $\mathcal{F}_2 = \{f : \mathbf{R} \to \mathbf{R} : f(v) = \mathbf{1}(v \leq t) : t \in \mathbf{R}\}$. It is routine to verify that $\mathcal{F}$ is a VC-subgraph class of functions with a measurable, square integrable envelope $\bar{f}(v) = \max(\max(r, v), 1)$, and that $\mathcal{F}$ is pointwise measurable and thus P-measurable (see Definition 2.3.3. and Example 2.3.4 in Van Der Vaart and Wellner, 1996). Thus, by Theorem 2.5.2 in Van Der Vaart and Wellner (1996), $\mathcal{F}$ is Donsker, so $\sqrt{n}(\hat{P} - P) \rightsquigarrow \mathbb{G}_P$ in $\ell^\infty(\mathcal{F})$, where $\hat{P} = n^{-1}\sum_{i=1}^n \delta_{S_i}$ is the empirical measure, and $\mathbb{G}_P$ is a tight centered Gaussian process with covariance kernel $K_P(f_1, f_2) = \mathbb{E}_P[f_1 f_2] - \mathbb{E}_P[f_1]\mathbb{E}_P[f_2]$.

First, consider estimation of the bounds on the expected profit. These bounds take the form

$$\pi(r; P, g) = \int \max(r, v) dP(v) - v_0 - (r - v_0) g(F_P(r)),$$

for continuously differentiable functions $g \in \{\underline{\psi}, \overline{\psi}\}$. Lemma D.1 below establishes that the map $P \mapsto \pi(\cdot; P, g)$ is Hadamard differentiable with the derivative

$$\pi'_{P,g}[H](r) = \int \max(r, v) dH(v) - g'(F_P(r)) F_H(r)(r - v_0). \tag{D.1}$$

Let

$$\hat{\pi}(r; g) = \pi(r; \hat{P}, g) = \frac{1}{n}\sum_{i=1}^n \max(r, S_i) - v_0 - (r - v_0) g(\hat{F}(r))$$

denote the plug-in estimator of profit bounds. By the functional delta method (Theorem 3.9.4. in Van Der Vaart and Wellner, 1996),

$$\sqrt{n}(\hat{\pi}(\,\cdot\,; g) - \pi(\,\cdot\,; P, g)) \overset{\ell^\infty(\mathcal{R})}{\rightsquigarrow} \pi'_{P,g}[\mathbb{G}_P](\cdot).$$

In particular, for each $r \in \mathcal{R}$, the above implies $\sqrt{n}(\hat{\pi}(r; g) - \pi(r; P, g)) \rightsquigarrow \mathcal{N}(0, V_{P,g})$,

where

$$V_{P,g} = \mathbb{V}ar_P(\max(r, S)) + (r - v_0)^2 g'(F_P(r))^2 F_P(r)(1 - F_P(r)) \\ - 2(r - v_0) g'(F_P(r))(r - \mathbb{E}_P[\max(r, S)]) F_P(r).$$

This expression can be used to construct pointwise confidence bands on the profit bounds.

Next, consider estimation of the MMR reserve price. The maximum regret takes the form $R(r; P) = \max\{R_1(r; P), R_2(r; P)\}$, where

$$R_1(r; P) = \max_{v_0 \le s \le r} \pi(s; P, g_{1,r}) - \pi(r; P, g_{1,r}),$$

$$R_2(r; P) = \max_{r \le s \le \overline{r}} \pi(s; P, g_{2,r}) - \pi(r; P, g_{2,r}).$$

The minimizer of $R(r; P)$ either minimizes $R_1(r; P)$ or $R_2(r; P)$, or solves $\Delta(r; P) = 0$, where $\Delta(r; P) = R_1(r; P) - R_2(r; P)$. For brevity, below we focus on the latter case since our empirically calibrated simulations suggest it is the most practically relevant. The arguments for the other two cases are very similar.

Let $s_1^*(r) = \text{argmax}_{v_0 \le s \le r} \pi(s; P, g_{1,r})$ and $s_2^*(r) = \text{argmax}_{r \le s \le \overline{r}} \pi(s; P, g_{2,r})$, which are unique, by Assumption D.1-(3). By Lemmas D.1 and D.2 and the chain rule (Lemma 3.9.3 in Van Der Vaart and Wellner, 1996), $P \mapsto \Delta(\cdot, P)$, viewed as a map from $\ell^\infty(\mathcal{F})$ to $\ell^\infty(\mathcal{R})$, is Hadamard differentiable at any P satisfying Assumption D.1-(3) tangentially to the set $\mathcal{H}_0 = \{H \in \ell^\infty(\mathcal{F}) : r \mapsto \pi'_{P,g}[H](r) \text{ is continuous}\}$, with the derivative

$$\Delta'_P[H](r) = \{\pi'_{P,g_{1,r}}[H](s_1^*(r)) - \pi'_{P,g_{1,r}}[H](r)\} - \{\pi'_{P,g_{2,r}}[H](s_2^*(r)) - \pi'_{P,g_{2,r}}[H](r)\},$$

where $\pi'_{P,g}[H](\cdot)$ are given in (D.1). Combining the above result with Lemma D.3, yields that $P \mapsto r^*(P)$, viewed as a map from $\ell^\infty(\mathcal{F})$ to $\mathbf{R}$, is Hadamard differentiable at any P satisfying conditions (3) and (4) of Assumption D.1, tangentially to $\mathcal{H}_0$ defined above, with the derivative

$$r'_P[H] = -\frac{\Delta'_P[H](r^*(P))}{\Delta'_r(r^*(P); P)}.$$

Thus, letting $\hat{r} = r^*(\hat{P}_n)$ denote any root of $\hat{\Delta}(r) = 0$ where $\hat{\Delta}(r) = \Delta(r; \hat{P}_n)$, by the

functional delta method, we obtain

$$\sqrt{n}(\hat{r} - r^*(P)) \quad \rightsquigarrow \quad r'_P[\mathbb{G}_P].$$

Bounds on the expected profit under MMR reserve immediately follow.

Finally, the validity of the proposed bootstrap procedure in Algorithm 1 follows from Theorems 3.6.3. and 3.9.11 in Van Der Vaart and Wellner (1996).

D.1 Auxiliary Lemmas

Lemma D.1. *Let $g : [0,1] \to [0,1]$ be continuously differentiable in the neighborhood of $cl(\{F_P(r) : r \in \mathcal{R}\})$. The map $\pi : \ell^\infty(\mathcal{F}) \to \ell^\infty(\mathcal{R})$ given by $P \mapsto \pi(\,\cdot\,; P, g)$ is Hadamard differentiable at P with the derivative*

$$\pi'_{P,g}[H](r) = \int \max(r, v) dH(v) - g'(F_P(r))F_H(r)(r - v_0).$$

Proof. Let $H_n \to H$ in $\ell^\infty(\mathcal{F})$ and $t_n \to 0$. By the mean-value theorem,

$$\frac{\pi(r; P + t_n H_n, g) - \pi(r; P, g)}{t_n} = \int \max(r, v) dH_n(v) - g'(\tilde{F}_n(r))F_{H_n}(r)(r - v_0),$$

where $\tilde{F}_n(r)$ is a convex combination of $F_{P+t_n H_n}(r)$ and $F_P(r)$. Thus, using the assumed continuity of $g'(\cdot)$ and convergence of $H_n \to H$ in $\ell^\infty(\mathcal{F})$,

$$\lim_{n\to\infty} \sup_{r\in\mathcal{R}} \left| \frac{\pi(r; P + t_n H_n, g) - \pi(r; P, g)}{t_n} - \pi'_{P,g}[H](r) \right| = 0.$$

■

Lemma D.2. *Let $\mathcal{R} = [v_0, \overline{r}]$, where $\overline{r} < \infty$. Define $\phi, \psi : \ell^\infty(\mathcal{R}) \to \ell^\infty(\mathcal{R})$ by*

$$\phi(g)(r) = \sup_{v_0 \le s \le r} g(s), \qquad \psi(g)(r) = \sup_{r \le s \le \overline{r}} g(s).$$

Let $f \in C(\mathcal{R})$ be strictly single-peaked, with unique maximizer r^*. Then ϕ and ψ are Hadamard differentiable at f, tangentially to $C(\mathcal{R})$, with derivatives

$$\phi'_f[h](r) = h(\min\{r, r^*\}), \qquad \psi'_f[h](r) = h(\max\{r, r^*\}),$$

for every $h \in C(\mathcal{R})$.

Proof. We prove the result for ϕ; the argument for ψ is analogous. Let $m(r) = \min\{r,r^*\}$, which is the unique maximizer of f over $[v_0,r]$.

Consider $t_n \downarrow 0$ and $h_n \to h$ uniformly, where $h \in C(\mathcal{R})$. For every $r \in \mathcal{R}$, choose an approximate maximizer $s_n(r) \in [v_0,r]$ such that

$$f(s_n(r)) + t_n h_n(s_n(r)) \geq \phi(f + t_n h_n)(r) - t_n^2.$$

Since $m(r)$ maximizes f over $[v_0,r]$,

$$0 \leq f(m(r)) - f(s_n(r)) \leq 2t_n \|h_n\|_{\mathcal{R}} + t_n^2.$$

We claim that

$$\sup_{r \in \mathcal{R}} |s_n(r) - m(r)| \to 0.$$

Indeed, for every $\varepsilon > 0$, strict single-peakedness and compactness imply

$$\inf_{\substack{r \in \mathcal{R},\ s \in [v_0,r] \\ |s - m(r)| \geq \varepsilon}} \{f(m(r)) - f(s)\} > 0,$$

whenever the indexing set is nonempty. The preceding inequality therefore establishes the claimed uniform convergence.

By the definition of $s_n(r)$,

$$h_n(m(r)) \leq \frac{\phi(f + t_n h_n)(r) - \phi(f)(r)}{t_n} \leq h_n(s_n(r)) + t_n.$$

Since $h_n \to h$ uniformly, h is uniformly continuous, and $s_n(r) \to m(r)$ uniformly in r, both bounds converge uniformly to $h(m(r))$. Consequently,

$$\left\| \frac{\phi(f + t_n h_n) - \phi(f)}{t_n} - h \circ m \right\|_{\infty} \to 0.$$

This establishes Hadamard differentiability of ϕ, with derivative $\phi'_f[h] = h \circ m$.

For ψ, the same argument applies with $m(r) = \max\{r,r^*\}$. ■

Lemma D.3. *Let the map* $\phi : \mathbb{D}_\phi \to \mathbf{R}$ *be defined as* $\phi(f)$ *equal to any point* $r \in \mathcal{R}$ *such that* $f(r) = 0$*, where* $\mathbb{D}_\phi = \{f \in \ell^\infty(\mathcal{R}) : f(r) = 0 \text{ for some } r \in \mathcal{R}\}$*. Let* $f \in \mathbb{D}_\phi$

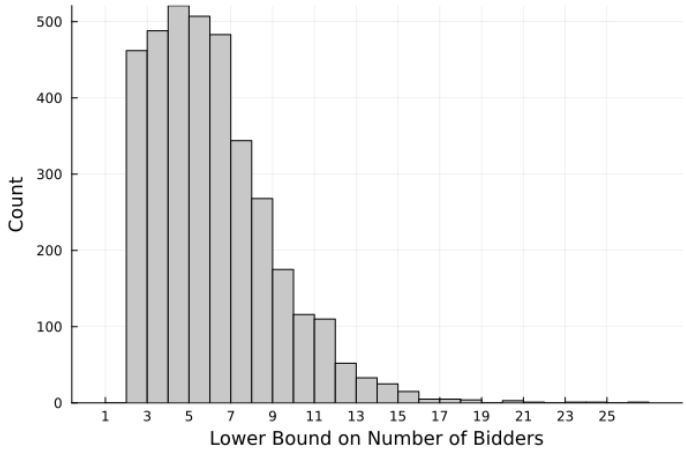


Figure 6: Distribution of the lower bound on number of bidders across all auctions.

be a continuous function such that $f(\tilde{r}) = 0 \iff \tilde{r} = r$*, continuously differentiable in the neighborhood of* r*, and satisfying* $f'(r) \neq 0$*. Then,* ϕ *is Hadamard differentiable at* f*, tangentially to the set of functions* $h \in \ell^\infty(\mathcal{R})$ *that are continuous at* r*, with the derivative*

$$\phi'_f[h] = -\frac{h(r)}{f'(r)}.$$

Proof. Let $h_n \to h$ in $\ell^\infty(\mathcal{R})$ and $t_n \to 0$. Denote $r_n = \phi(f + t_n h_n)$ and $r = \phi(f)$. Then

$$(f + t_n h_n)(r_n) = f(r) \quad \Longrightarrow \quad |f(r_n) - f(r)| = t_n |h_n(r_n)|.$$

Since $h_n(r) \to h(r)$ uniformly over $r \in \mathcal{R}$, and $t_n \to 0$, we have $|f(r_n) - f(r)| \to 0$. Since r is the unique point at which $f(r) = 0$, and $f'(r) \neq 0$, the above implies that $r_n \to r$. Thus, using the above display and the mean-value theorem, we can express

$$\frac{f(r_n) - f(r)}{t_n} = -h_n(r_n) \implies \frac{r_n - r}{t_n} = -\frac{h_n(r_n)}{f'(\tilde{r}_n)},$$

for some $\tilde{r}_n$ between r_n and r. Since $h_n(r) \to h(r)$ uniformly over $r \in \mathcal{R}$, and h is continuous at r, we have $h_n(r_n) = h(r_n) + o(1) = h(r) + o(1)$. Since $f'(\cdot)$ is continuous around r, $f'(r) \neq 0$, and $\tilde{r}_n \to r$, we have $f'(\tilde{r}_n) \to f'(r)$, and the result follows. ■

E Additional Tables and Figures

Saleroom Location	Christie's		Sotheby's	
	Threshold	Rate	Threshold	Rate
Hong Kong	≤HK$7.5M	26.0%	≤HK$7,500,000	26.0%
	>HK$7.5M and ≤ HK$50M	20.0%	>HK$7.5M and ≤HK$40M	20.0%
	>HK$50M	14.5%	>HK$40M	13.9%
London	≤700k	26.0%	≤800k	26.0%
	>700,000 and ≤4.5M	20.0%	>800k and ≤3.8M	20.0%
	>4.5M	14.5%	>3.8M	13.9%
Paris	≤700k	26.0%	≤800k	26.0%
	>700,000 and ≤4M	20.0%	>800k and ≤3.5M	20.0%
	>4M	14.5%	>3.5M	13.9%
New York	≤$1M	26.0%	≤$1M	26.0%
	>$1M and ≤$6M	20.0%	>$1M and ≤$4.5M	20.0%
	>$6M	14.5%	>$4.5M	13.9%
Shanghai	≤6M	26.0%	—	—
	>6M and ≤40M	20.0%	—	—
	>40M	14.5%	—	—

This table is accurate as of February 7 2022 for Christie's and February 1 2023 for Sotheby's. In the last 10 years, there are only minor changes to the base rate (i.e. lowest threshold category). These buyer premium thresholds are additive, so final transaction amounts are strictly increasing. *Source:* Christie's and Sotheby's Websites.

Table 6: Buyer's Premiums in Christie's and Sotheby's Auctions

High Estimate Range	[$100K, $1.00M]			[$1.00M, $10.00M]		
Number of Bidders	$N \in [2,10]$	$N \in [6,40]$	$N \in [2,40]$	$N \in [2,10]$	$N \in [6,40]$	$N \in [2,40]$
Avg High Estimate	$532,443	$483,529	$507,771	$3,761,415	$3,641,264	$3,696,885
Avg Realized Profit	0.75	1.65	1.24	0.31	0.95	0.62
$[\underline{\rho}, \overline{\rho}]$	[0.70, 0.70]	[0.40, 0.40]	[0.50, 0.50]	[0.60, 0.60]	[0.65, 0.65]	[0.55, 0.55]
Convex MMR reserve	0.96	1.36	1.54	0.90	0.86	1.03
– 95% CI (reserve)	[0.81, 1.38]	[1.11, 1.71]	[1.35, 1.66]	[0.79, 1.12]	[0.83, 0.89]	[0.84, 1.52]
Bounds on profit	[0.76, 0.80]	[1.67, 1.70]	[1.21, 1.36]	[0.32, 0.36]	[0.96, 0.96]	[0.62, 0.68]
– 95% CI (profit)	[0.59, 0.97]	[1.51, 1.88]	[1.10, 1.47]	[0.27, 0.42]	[0.87, 1.04]	[0.52, 0.80]
Max-min reserve	0.81	1.19	0.99	0.77	0.86	0.82
– 95% CI (reserve)	[0.79, 0.85]	[1.11, 1.28]	[0.95, 1.03]	[0.75, 0.78]	[0.83, 0.88]	[0.81, 0.85]
Bounds on profit	[0.77, 0.78]	[1.68, 1.69]	[1.26, 1.28]	[0.33, 0.34]	[0.96, 0.96]	[0.64, 0.65]
– 95% CI (profit)	[0.65, 0.91]	[1.52, 1.86]	[1.16, 1.39]	[0.30, 0.38]	[0.87, 1.04]	[0.59, 0.70]
$[\underline{\rho}, \overline{\rho}]$	[0.70, 0.80]	[0.40, 0.50]	[0.50, 0.60]	[0.60, 0.70]	[0.65, 0.75]	[0.55, 0.65]
Convex MMR reserve	1.17	1.24	1.36	0.84	0.81	0.93
– 95% CI (reserve)	[0.77, 1.21]	[1.03, 1.55]	[1.19, 1.47]	[0.75, 1.04]	[0.79, 0.86]	[0.84, 1.37]
Bounds on profit	[0.71, 0.82]	[1.67, 1.69]	[1.22, 1.34]	[0.32, 0.35]	[0.95, 0.96]	[0.63, 0.67]
– 95% CI (profit)	[0.61, 0.95]	[1.51, 1.87]	[1.11, 1.44]	[0.28, 0.42]	[0.87, 1.05]	[0.53, 0.78]
Max-min reserve	0.78	1.10	0.92	0.75	0.81	0.79
– 95% CI (reserve)	[0.75, 0.80]	[1.03, 1.17]	[0.89, 0.95]	[0.74, 0.76]	[0.79, 0.83]	[0.78, 0.80]
Bounds on profit	[0.77, 0.77]	[1.67, 1.68]	[1.26, 1.27]	[0.33, 0.34]	[0.95, 0.96]	[0.63, 0.64]
– 95% CI (profit)	[0.65, 0.90]	[1.51, 1.85]	[1.15, 1.38]	[0.30, 0.37]	[0.87, 1.04]	[0.59, 0.69]
$[\underline{\rho}, \overline{\rho}]$	[0.00, 1.00]	[0.00, 1.00]	[0.00, 1.00]	[0.00, 1.00]	[0.00, 1.00]	[0.00, 1.00]
Reserve*	0.67	0.67	0.67	0.67	0.67	0.67
– 95% CI (reserve)	[0.67, 0.67]	[0.67, 0.67]	[0.67, 0.67]	[0.67, 0.68]	[0.67, 0.67]	[0.67, 0.67]
Bounds on profit	[0.76, 0.76]	[1.66, 1.66]	[1.25, 1.25]	[0.33, 0.33]	[0.95, 0.95]	[0.63, 0.63]
– 95% CI (profit)	[0.65, 0.89]	[1.51, 1.83]	[1.15, 1.37]	[0.29, 0.37]	[0.87, 1.04]	[0.59, 0.68]
Observations	131	153	314	311	269	610

*For the unrestricted range $\rho \in [0,1]$, the convex MMR and max-min reserves (and their profit bounds) coincide.

Table 7: Convex minimax-regret and max-min reserves with profit bounds (and bootstrap 95% CIs) for Modern Art, New York (scaled by high estimate).